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Causal Identification with Matrix Equations

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Abstract

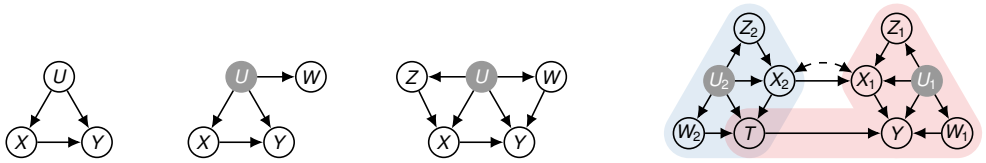
Causal effect identification is concerned with determining whether a causal effect is computable from a combination of qualitative assumptions about the underlying system (e.g., a causal graph) and distributions collected from this system. Many identification algorithms exclusively rely on graphical criteria made of a non-trivial combination of probability axioms, do-calculus, and refined c-factorization (e.g., Lee & Bareinboim, 2020). In a sequence of increasingly sophisticated results, it has been shown how *proxy* variables can be used to identify certain effects that would not be otherwise recoverable in challenging scenarios through solving matrix equations (e.g., Kuroki & Pearl, 2014; Miao et al., 2018). In this paper, we develop a new causal identification algorithm which utilizes both graphical criteria and matrix equations. Specifically, we first characterize the relationships between certain graphically-driven formulae and matrix multiplications. With such characterizations, we broaden the spectrum of proxy variable based identification conditions and further propose novel *intermediary* criteria based on the pseudoinverse of a matrix. Finally, we devise a causal effect identification algorithm, which accepts as input a collection of marginal, conditional, and interventional distributions, integrating enriched matrix-based criteria into a graphical identification approach.

1 Introduction

Cause and effect relations are one of the most common types of knowledge sought after throughout the empirical sciences. These relations are one of the main ingredients in the construction of stable explanations, and usually underpin robust and generalizable decision-making strategies [18, 24]. There is a growing literature that aims to systematically find causal relations by fusing observations, experiments, and substantive knowledge about the phenomenon under investigation [18, 3, 19]. Formally, the inferential target usually appears as the effect of a set of variables $do(\mathbf{X} = \mathbf{x})$ on another set of variables $\mathbf{Y}$, which is written as $P(\mathbf{y}|do(\mathbf{x}))$ or $P_{\mathbf{x}}(\mathbf{y})$. Assumptions about the underlying data-generating processes are commonly expressed as a causal graph $\mathcal{G}$ over endogenous variables $\mathbf{V}$. There are different lines of investigation that aims to establish the quantity $P_{\mathbf{x}}(\mathbf{y})$.

First, one line of investigation attempts to exploit the non-parametric constraints encoded in $\mathcal{G}$ to determine whether the quantity $P_{\mathbf{x}}(\mathbf{y})$ (i.e., query) is uniquely computable from the different available distributions. For instance, Pearl's celebrated method known as *do-calculus* provides a symbolic way of determining whether a query can be identified from $\mathcal{G}$ and observational data $P(\mathbf{V})$ [17]. A number of necessary and sufficient conditions were developed for systematically determining the identifiability status of the query from observational data [17, 27, 21, 9]. For example, the causal effect $P_x(y)$ in Fig. 1a is identified by a back-door criterion as $P_x(y) = \sum_u P(y|x,u)P(u)$. Further methods were developed to identify the query from a combination of observational and experimental distributions [2, 13, 12, 11]. The key ideas behind these methods are (i) to decompose the given

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(a) (b) (c) (d) $\mathcal{G}$

Figure 1: Causal graphs: (a) with a back-door condition; (b) with W as a proxy for U; (c) with proxies W and Z for U. (d) A causal graph $\mathcal{G}$ where a causal effect $P_{\mathbf{x}}(y)$ can be computed by estimating each of $P_{x_2}(t)$ (light blue, left) and $P_{t,x_1}(y)$ (light red, right) using MGT criterion.

causal query into standard *factors* (following Tian's *c-factorization*, [27]), leveraging the graphical constraints, and (ii) to identify these factors individually, matching one of the available distributions. We call this the factorization-based approach.

Another line of investigation attempts to exploit assumptions about the relationship between unobserved confounders and the observable variables through the idea of proxy variables [5, 8, 10, 14, 30]. These methods rely on the cardinality and complexity of these relationships, which will possibly lead to the invertibility of certain matrices. The focus is usually on local conditions involving the treatment X, the outcome Y, and the proxies W, Z for the set of unobserved confounders U; for example, see Fig. 1b. Note that since U is unobserved, the effect of X on Y, i.e., $P_x(y)$, is provably not identifiable by standard graphical methods discussed earlier. Still, if the distribution of the proxy given the unobserved confounder is available $P(W|U)$, in addition to $P(X, Y, W)$, the quantity $P_x(y)$ can be computed under the invertibility of $\mathbf{P}(W|U)$, a matrix representation of $P(W|U)$ where $\mathbf{P}(W|U)_{i,j} = P(w_i|u_j)$ (see Eq. (3) for detail). In case of Fig. 1c, the causal effect $P_x(y)$ is identified from $P(X, Y, W, Z)$ through the following,[1]

$$P_x(y) = \mathbf{P}(y|Z, x)\mathbf{P}(W|Z, x)^{\dagger}\mathbf{P}(W), \tag{1}$$

where $\mathbf{P}(W|Z, x)$ has rank matching the size of domain of the unmeasured confounder U. Although typically not framed in terms of identification with multiple datasets, this setting corresponds to identifying $P_x(y)$ with, e.g., marginal distributions $\{P(X, Y, Z), P(W, X, Z)\}$ or conditional distributions $\{P(Y|Z, x), P(W|Z, x), P(W)\}$. We call this the proxy approach.[2]

Despite all the power and successes achieved by the factorization and proxy approaches, there exist still interesting cases not covered by any of them individually. To witness, consider the causal graph in Fig. 1d and first notice that the effect is *not* identified from each approach. On the other hand, if we combine both approaches, the causal effect is identifiable and given by,

$$\begin{aligned} P_{\mathbf{x}}(y) &= \textstyle\sum_t P_{\mathbf{x}}(t, y) = \sum_t P_{t,x_1}(y)P_{x_2}(t) \\ &= \textstyle\sum_t \left[\mathbf{P}(y|Z_1, x_1, t)\mathbf{P}(W_1|Z_1, x_1, t)^{\dagger}\mathbf{P}(W_1)\right]\left[\mathbf{P}(t|Z_2, x_2)\mathbf{P}(W_2|Z_2, x_2)^{\dagger}\mathbf{P}(W_2)\right], \end{aligned}$$

where $P_{t,x_1}(y)$ and $P_{x_2}(t)$, the two factors of the query, are individually identified with the help of the proxy approach, i.e., Eq. (1).

Our goal in this paper is to explicate how this can be accomplished from first principles. More broadly, we will develop a flexible and general graphical identification approach that combines both the factorization and matrix equations (the underpinning idea of proxy-based methods). More specifically, our contributions are as follows:

1. We connect the graphical and matrical approaches by characterizing matrix equations of probability distributions driven by graphical constraints in a causal graph. This leads to a better understanding of the identification through solving systems of equations.

[1] We focus on finite discrete variables. Detailed discussion on a continuous case can be found in [14]. We refer this identification condition *MGT criterion* where the acronym MGT comes from the surnames of the authors in [14].

[2] Recent work in this line of research use the term *proximal causal inference* (PCI) [23, 25, 31, 6]. [25] generalizes the results in [14] to allow observed confounders as well. [23] considered combining graphical and proximal approaches, similar to the motivational example.

2. Building on this new characterization, we generalize proxy-based criteria and devise novel intermediary pseudoinverse criteria so as to identify a causal effect by utilizing the general inverse of a matrix and diverse collection of distributions.
3. We develop a general identification algorithm that amalgamates graphical and matrical approaches, returning an identification formula for a causal query given a causal diagram and a set of marginal, experimental, and conditional distributions. We show that this method subsumes current state of the art in the literature.

2 Preliminaries

We follow notational conventions from literature on causal inference. We denote a variable by an upper case letter Y, and its value is denoted by its corresponding lower case letter y in the domain $\mathfrak{X}_Y$. A set of variables will be denoted by a bold capital letter $\mathbf{Y}$ with its value $\mathbf{y}$. We may use $\dot\cup$, instead of $\cup$, to emphasize the union of two *disjoint* sets. Given $\mathbf{Z} \subseteq \mathbf{W}$, $\mathbf{w}\backslash\mathbf{Z}$ denotes the value of $\mathbf{W}\backslash\mathbf{Z}$ consistent with $\mathbf{w}$. Let $\mathbf{a}/\mathbf{B} = (\mathbf{A} \cap \mathbf{B}, \mathbf{a}\backslash\mathbf{B})$ which retains $\mathbf{A} \cap \mathbf{B}$ as variables and fixes the remaining variables of $\mathbf{A}$ at their values in $\mathbf{a}$. Without loss of generality, we refer $P_{\mathbf{Z}}(\mathbf{V}'|\mathbf{W})$ a distribution. We may employ *conditional* to emphasize $\mathbf{W} \neq \emptyset$, *experimental* or *interventional* $\mathbf{Z} \neq \emptyset$ compared to *observational* $\mathbf{Z} = \emptyset$, and *marginal* if $\mathbf{Z} \cup \mathbf{W} \cup \mathbf{V}' \subsetneq \mathbf{V}$.

All conditional probabilities used below are evaluated on their conditioning support. Elementwise divisions are subject to the overlap condition stated with the relevant criterion.

We employ structural causal models (SCMs) [18, Ch. 7] as the semantical framework to represent a domain of interest. An SCM $\mathcal{M}$ is a quadruple $\langle \mathbf{U}, \mathbf{V}, P(\mathbf{U}), \mathbf{F}\rangle$. A set of exogenous variables $\mathbf{U}$, which follows $P(\mathbf{U})$, is determined by factors outside the model. $\mathbf{V}$ is a set of endogenous variables whose values are determined by functions $\mathbf{F} = \{f_i\}_{V_i \in \mathbf{V}}$ such that $V_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i)$ where $\mathbf{PA}_i \subseteq \mathbf{V}\backslash\{V_i\}$ and $\mathbf{U}_i \subseteq \mathbf{U}$. Further, $do(\mathbf{x})$ represents the operation of holding a set $\mathbf{X}$ to a constant $\mathbf{x}$ regardless of their original mechanisms. Such intervention induces a submodel $\mathcal{M}_{\mathbf{x}}$, which is $\mathcal{M}$ with f_X replaced to x for $X \in \mathbf{X}$. The distribution over $\mathbf{V}$ induced by the submodel is denoted by $P_{\mathbf{x}}(\mathbf{V})$. We may employ letter Q to denote an interventional distribution, e.g., $Q = P_{\mathbf{r}}$.

Each SCM (model, for short) induces a causal diagram (or causal graph) $\mathcal{G} = \langle \mathbf{V}, \mathbf{E}\rangle$, where each type of edge represents a different causal relationship among the variables: (i) $X \rightarrow Y$ if X is used as an argument of f_Y; and (ii) $X \leftrightarrow Y$ if $\mathbf{U}_X$ and $\mathbf{U}_Y$ are correlated. Given a causal diagram $\mathcal{G}$, familial relationships among its vertices are denoted by pa and an for parents and ancestors, respectively. Further, An is a set of ancestors including its argument as well. We denote by $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ an edge subgraph of $\mathcal{G}$ which removes edges incoming to $\mathbf{X}$ and outgoing from $\mathbf{Z}$. Causal relationships among other variables are captured in $\mathcal{G}\backslash\mathbf{X}$, which is the subgraph of $\mathcal{G}$ over $\mathbf{V}\backslash\mathbf{X}$. A vertex induced subgraph is denoted by $\mathcal{G}[\mathbf{V}']$ where $\mathbf{V}' \subseteq \mathbf{V}$. Causal effect identification relies heavily on standard graphical constraints imposed by a causal diagram such as d-separation (reading off conditional independence from the graph, [28, 7]) and do-calculus (equivalence among interventional probabilities) [17], which we present in the Appendix for completeness.

The *latent projection* (or projection, for short) of a causal diagram is a causal diagram retaining the causal relationships among a subset of variables. We denote by $\mathcal{G}\langle\mathbf{V}'\rangle$ the latent projection of $\mathcal{G}$ onto $\mathbf{V}' \subseteq \mathbf{V}$, the causal graph over $\mathbf{V}'$ [29]. Conditional independence (CI) statements and do-calculus [17] on a projection are valid in $\mathcal{G}$, vice versa. We formally define a latent projection in the supplementary material. The omitted proofs and derivations are also provided in the appendix.

3 Characterization of Matrix Equations of Graphical Constraints

In this section, we present characterizations of graphical constraints in a given causal diagram $\mathcal{G}$, leading to equations expressed as the multiplication of matrices. The characterizations will further advance our understanding on the constraints imposed over the distributions generated by the underlying system compared to simple equivalence relationships such as conditional independence and do-calculus. To begin with, we denote by $\mathbf{P}$ the matrix representation of a distribution P where free outcome variables are rows and free condition or intervention variables correspond to columns. For instance, $\mathbf{P}_{a,B}(C, d)$ is a $|\mathfrak{X}_C| \times |\mathfrak{X}_B|$ matrix and $\mathbf{P}_r(A, b|C, d)$ is a $|\mathfrak{X}_A| \times |\mathfrak{X}_C|$ matrix. Further, we may use $'$ and $''$ to represent two disjoint subsets such that $\mathbf{B} = \mathbf{B}' \dot\cup \mathbf{B}''$.

Chain Rule with Conditional Independence The definitions of conditional and marginal distributions naturally lead to a sum of a product of probabilities. Let $Q = P_{\mathbf{r}}$ be an arbitrary interventional distribution. Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{R}$ be disjoint. A marginal probability over chain-rule-induced multiplication is expressed as

$$Q(\mathbf{a}, \mathbf{b}'|\mathbf{c}) = \sum_{\mathbf{b}''} Q(\mathbf{a}|\mathbf{b}, \mathbf{c})Q(\mathbf{b}|\mathbf{c}) = \mathbf{Q}(\mathbf{a}|\mathbf{b}', \mathbf{B}'', \mathbf{c})\mathbf{Q}(\mathbf{B}'', \mathbf{b}'|\mathbf{c}).$$

Considering conditional independence, we can further enrich such a characterization.

Lemma 1. *Given a causal diagram $\mathcal{G}$, let $Q = P_{\mathbf{r}}$ for some $\mathbf{r} \in \mathfrak{X}_{\mathbf{R}}$ where $\mathbf{R} \subsetneq \mathbf{V}$. Let $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}$ be disjoint subsets of $\mathbf{V}\backslash\mathbf{R}$. If $(\mathbf{D} \perp\!\!\!\perp \mathbf{A} \mid \mathbf{B}, \mathbf{C}, \mathbf{E})$ and $(\mathbf{E} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}, \mathbf{D})$ in $\mathcal{G}\backslash\mathbf{R}$, then, $\mathbf{Q}(\mathbf{A}, \mathbf{b}'|\mathbf{c}, \mathbf{D}, \mathbf{e}) = \mathbf{Q}(\mathbf{A}|\mathbf{b}', \mathbf{B}'', \mathbf{c}, \mathbf{e})\mathbf{Q}(\mathbf{B}'', \mathbf{b}'|\mathbf{c}, \mathbf{D})$.*

The lemma emphasizes the condition under which the result of multiplication is a matrix not just a row or column. A special case of the lemma appears in the MGT criterion where the inverse of a matrix multiplication $\mathbf{P}(W|Z, x) = \mathbf{P}(W|U, x)\mathbf{P}(U|Z, x)$ is utilized so as to cancel out an unknown distribution $\mathbf{P}(W|U) = \mathbf{P}(W|U, x)$.

Adjustment Criterion Given a graph $\mathcal{G}$ and a causal effect of interest $P_{\mathbf{x}}(\mathbf{y})$, the adjustment criterion [22] seeks a set of covariates $\mathbf{Z} \subseteq \mathbf{V}\backslash\mathbf{X}\backslash\mathbf{Y}$, called an *adjustment set* for a causal effect $P_{\mathbf{x}}(\mathbf{y})$, which grants the following expression, $P_{\mathbf{x}}(\mathbf{y}) = \sum_{\mathbf{z}} P(\mathbf{y}|\mathbf{x}, \mathbf{z})P(\mathbf{z})$. Adjustment criterion generalizes back-door criterion [18]. Its matricized expression with employing $Q = P_{\mathbf{r}}$ is

$$Q_{\mathbf{x}}(\mathbf{y}) = \sum_{\mathbf{z}} Q(\mathbf{y}|\mathbf{x}, \mathbf{z})Q(\mathbf{z}) = \mathbf{Q}(\mathbf{y}|\mathbf{x}, \mathbf{Z})\mathbf{Q}(\mathbf{Z}).$$

This simple expression plays a central role in the identification with proxy variables.

In many settings, the left hand side (LHS) is the query of interest and two terms in the RHS are usually available or to be inferred using other available quantities. However, substituting value $\mathbf{y}$ with $\mathbf{Y}$, we can further yield (under invertibility assumption) $\mathbf{Q}(\mathbf{Z}) = \mathbf{Q}(\mathbf{Y}|\mathbf{x}, \mathbf{Z})^{-1}\mathbf{Q}_{\mathbf{x}}(\mathbf{Y})$, which restores the covariate distribution of interest given a causal effect and conditional distribution.

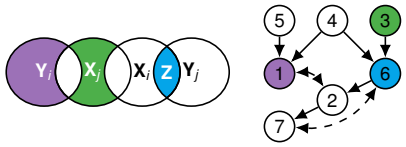
C-Factorization C-factorization [26] decomposes a causal effect $P_{\mathbf{x}}(\mathbf{y})$ into the sum-product of c-factors (simply, factors) with respect to the given causal diagram $\mathcal{G}$. Without loss of generality, let $\mathbf{X}$ be minimal such that no $\mathbf{X}' \subsetneq \mathbf{X}$ satisfies $P_{\mathbf{x}'}(\mathbf{y}) = P_{\mathbf{x}}(\mathbf{y})$ (i.e., overriding $\mathbf{X}$ by $\mathrm{an}^{\mathcal{G}_{\overline{\mathbf{x}}}}(\mathbf{Y}) \cap \mathbf{X}$). For any projection $\mathcal{H}$ of $\mathcal{G}$ that preserves $\mathbf{X} \cup \mathbf{Y}$, the following holds:

$$P_{\mathbf{x}}(\mathbf{y}) = \sum_{\mathbf{y}^+\backslash\mathbf{y}} P_{\mathbf{x}}(\mathbf{y}^+) = \sum_{\mathbf{y}^+\backslash\mathbf{y}} \prod_{\mathbf{Y}_i \in \mathcal{C}(\mathcal{H}[\mathbf{Y}^+])} P_{\mathrm{pa}^{\mathcal{H}}(\mathbf{y}_i)\backslash\mathbf{y}_i}(\mathbf{y}_i), \tag{2}$$

where $\mathbf{Y}^+ = \mathrm{An}^{\mathcal{H}_{\underline{\mathbf{x}}}}(\mathbf{Y})$ and $\mathcal{C}(\cdot)$ is the c-component decomposition (partitioning the variables in the graph based on their connectivity through bidirected edges). Given a query $P_{\mathbf{x}}(\mathbf{y})$, we let c-factors $\mathbb{F}_{\mathcal{H}} = \{\langle \mathbf{X}_i, \mathbf{Y}_i \rangle\}_i$ where $\{\mathbf{Y}_i\}_i$ form the c-component decomposition of $\mathcal{H}[\mathbf{Y}^+]$ and $\mathbf{X}_i = \mathrm{pa}^{\mathcal{H}}(\mathbf{Y}_i)\backslash\mathbf{Y}_i$. The general identification method under partial-observability [12] can be summarily described as finding $\mathcal{H}$ such that each factor is identified by one of the available distributions.

Let us denote $\mathbf{X}_{ij} = \mathbf{X}_i \cup \mathbf{X}_j$. We focus here on the sum-product of a pair of factors at some latent projection $\mathcal{H}$, which satisfies $P_{\mathbf{x}_i}(\mathbf{y}_i)P_{\mathbf{x}_j}(\mathbf{y}_j) = P_{\mathbf{x}_{ij}\backslash\mathbf{y}_{ij}}(\mathbf{y}_{ij})$ where the values are consistent [12]. Now consider the summation over $\mathbf{Z} \subseteq \mathbf{X}_i \cap \mathbf{Y}_j$. Then,

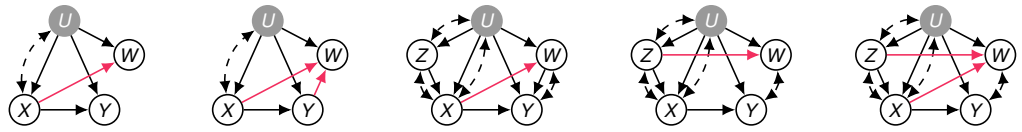
$$\sum_{\mathbf{z}} P_{\mathbf{x}_i\backslash\mathbf{z},\mathbf{z}}(\mathbf{y}_i)P_{\mathbf{x}_j}(\mathbf{y}_j\backslash\mathbf{z}, \mathbf{z}) = P_{\mathbf{x}_{ij}\backslash\mathbf{y}_{ij}}(\mathbf{y}_{ij}\backslash\mathbf{Z}).$$



(a) Euler diagram (b)

Figure 2: (a) set relationships, (b) a causal graph where the numbers correspond to the regions of (a) from the left to right.

To properly represent this as a matrix multiplication, we should decide which variables will become rows and columns. Two matrices to be multiplied should share only $\mathbf{Z}$, and other common variables, appearing in the two terms, or non-matching variables, appearing as intervention in left term and outcome in right term, need to be set to constants. To help understand, we illustrate in Fig. 2a the set relationships among $\mathbf{X}_i$, $\mathbf{Y}_i$, $\mathbf{X}_j$, and $\mathbf{Y}_j$ in the case of $\mathbf{Z} = \mathbf{X}_i \cap \mathbf{Y}_j$.



(a) $P_x(W)$ (b) $P_{x,y}(W)$ (c) $P_x(W)$ (d) $P_z(W)$ (e) $P_{x,z}(W)$

Figure 3: Causal diagrams admitting single- and double-proxy settings with surrogate experiments.

Therefore, $\mathbf{Y}_i \cap \mathbf{X}_j$ (the shared variables other than $\mathbf{Z}$) and $(\mathbf{X}_i \cup \mathbf{Y}_j) \backslash \mathbf{Z}$ (the non-matching variables) are fixed. These fixed variables are not colored in Fig. 2a. As a result, we can obtain a submatrix of $\mathbf{P}_{\mathbf{X}_{ij} \backslash \mathbf{Y}_{ij}}(\mathbf{Y}_{ij} \backslash \mathbf{Z})$ as the multiplication of the submatrices of $\mathbf{P}_{\mathbf{X}_i}(\mathbf{Y}_i)$ and $\mathbf{P}_{\mathbf{X}_j}(\mathbf{Y}_j)$ obtained via fixing variables.

Lemma 2 (Matrix Equation of C-Factorization with Two Factors)**.** *Given a causal diagram $\mathcal{G}$ and an experimental distribution $Q = P_{\mathbf{r}}$ where a causal effect is c-factorized as $Q_{\mathbf{x}}(\mathbf{y}) = \sum_{\mathbf{z}} Q_{\mathbf{x}_i}(\mathbf{y}_i) Q_{\mathbf{x}_j}(\mathbf{y}_j)$ in a projection $\mathcal{H}$ of $\mathcal{G} \backslash \mathbf{R}$, the effect can be represented as a matrix multiplication, if $\mathbf{Z} \subseteq \mathbf{X}_i \cap \mathbf{Y}_j$. Further, the corresponding matrix equation is*

$$\mathbf{Q}_{(\mathbf{x}_{ij} \backslash \mathbf{y}_{ij})/(\mathbf{X}_j \backslash \mathbf{X}_i \backslash \mathbf{Y}_i)}((\mathbf{y}_{ij} \backslash \mathbf{Z})/(\mathbf{Y}_i \backslash \mathbf{X}_j)) = \mathbf{Q}_{\mathbf{x}_i/\mathbf{Z}}(\mathbf{y}_i/(\mathbf{Y}_i \backslash \mathbf{X}_j)) \mathbf{Q}_{\mathbf{x}_j/(\mathbf{X}_j \backslash \mathbf{X}_i \backslash \mathbf{Y}_i)}(\mathbf{y}_j/\mathbf{Z}).$$

We illustrate a causal graph in Fig. 2b where each variable matches to each region in the Euler diagram (Fig. 2a). With $\mathbf{Z} = \{6\}$ and for an arbitrary instantiation of variables V_2, V_4, V_5, and V_7, $\mathbf{P}_{V_3,v_4,v_5}(V_1, v_2, v_7) = \mathbf{P}_{v_4,v_5,V_6}(V_1, v_2) \mathbf{P}_{v_2,V_3,v_4}(V_6, v_7)$.

In this section, we connected different graphical constraints—from chain-rule with conditional independence to c-factorization—induced identification formulae to matrix equations. Results presented in this section are by no means complete yet cover, to the best of our knowledge, every sum-product expression appeared in both graphical and matrical identification approaches. Nevertheless, this suite of characterizations will provide a fundamental understanding of the mathematical structures involved in the identification methods with matrical expressions.

4 Generalized Proxy-based Criteria

Equipped with the characterization from the previous section, we revisit single- and double-proxy settings [5, 8, 10, 14] more formally, which identify a causal effect through the combination of chain rule, adjustment criterion (c-factorization), and inverses of matrices given an observational distribution and additional external study involving an unobserved confounder. We investigate its extension capable of utilizing other types of available distributions. Results presented in this section is crucial in adopting matrical approaches into a factorization-based identification algorithm.

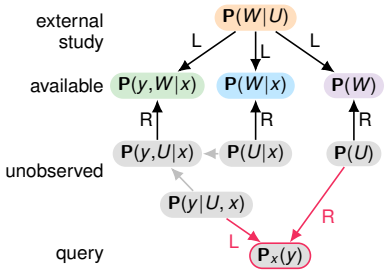


Figure 4: Schematic for a single proxy setting. Gray and red lines are for elementwise matrix multiplication and adjustment criterion, respectively.

Single-Proxy Setting We illustrate in Fig. 4 the available distributions and unknown distributions, considered in Fig. 1b, that can lead to $P_x(y)$. In the figure, a matrix multiplication $A = B \cdot C$ is represented as $B \xrightarrow{\mathsf{L}} A \xleftarrow{\mathsf{R}} C$ with positions annotated. First, note that $(X, Y \perp\!\!\!\perp W \mid U)$ in a causal graph $\mathcal{G}$ is central, which grants $P(W|U, x, y) = P(W|U, x) = P(W|U)$. If $\mathbf{P}(W|U)$ is invertible,[3] the three distributions $\mathbf{P}(y, U|x)$, $\mathbf{P}(U|x)$, and $\mathbf{P}(U)$ are obtained where $\mathbf{P}(y|U, x)$ is computed by chain rule (Lemma 1). Then, $P_x(y)$ is, with $\oslash$ denoting elementwise division,

$$\Big(\overbrace{\big(\underbrace{(\mathbf{P}(W|U)^{-1} \mathbf{P}(y, W|x))}_{\mathbf{P}(y,U|x)} \oslash \underbrace{(\mathbf{P}(W|U)^{-1} \mathbf{P}(W|x))}_{\mathbf{P}(U|x)} \big)^{\top}}^{\mathbf{P}(y|U,x)} \underbrace{\mathbf{P}(W|U)^{-1} \mathbf{P}(W)}_{\mathbf{P}(U)} . \tag{3}$$

[3] In case of W has more categorical values than U, one can coarsen W to W' ensuring that $\mathbf{P}(W'|U)$ is full rank. In other words, $\mathbf{P}(W|U)$ has full column rank.

The elementwise division in Eq. (3) is evaluated only for u with $P(u) > 0$, with $P(u|x) > 0$ on that support.

However, this scheme does not directly apply to Fig. 3a or 3b. When only Y is independent of W given U and X, the external study $P(W|U, x)$, surrogate experiment $P_x(W)$, and observational distribution $P(Y, W, X)$ suffice for Fig. 3a. For Fig. 3b, the stronger external study $P(W|U, x, Y)$ and surrogate experiment $P_{x,y}(W)$ are required by the derivation (see the appendix for both derivations). These examples suggest that we can generalize the single proxy-based criterion to take advantage of a diverse collection of external studies and surrogate experiments.

Theorem 1. *Given $\mathcal{G}$, let $\mathbf{X}$, $\mathbf{Y}$, $\mathbf{W}$, $\mathbf{U}$, and $\mathbf{R}$ be disjoint subsets of $\mathbf{V}$, $Q = P_{\mathbf{r}}$ for some $\mathbf{r} \in \mathfrak{X}_{\mathbf{R}}$, and $\mathcal{H} = \mathcal{G}\backslash\mathbf{R}$. A causal effect $P_{\mathbf{r},\mathbf{x}}(\mathbf{y}) = Q_{\mathbf{x}}(\mathbf{y})$ is identifiable if (1) $\mathbf{U}$ is an adjustment admissible set for $Q_{\mathbf{x}}(\mathbf{y})$ in $\mathcal{H}$, and $Q(\mathbf{u}|\mathbf{x}) > 0$ for every $\mathbf{u}$ with $Q(\mathbf{u}) > 0$; (2a) $(\mathbf{Y} \perp\!\!\!\perp \mathbf{W} \mid \mathbf{U}, \mathbf{X})_{\mathcal{H}}$ and $Q(\mathbf{W}|\mathbf{U}, \mathbf{x})$, $Q(\mathbf{y}, \mathbf{W}|\mathbf{x})$ and $Q(\mathbf{W}|\mathbf{x})$ are available where $\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x})$ has full column rank; or (2b) $(\mathbf{Y} \not\perp\!\!\!\perp \mathbf{W} \mid \mathbf{U}, \mathbf{X})_{\mathcal{H}}$ and $Q(\mathbf{W}|\mathbf{U}, \mathbf{x}, \mathbf{Y})$ and $Q(\mathbf{Y}, \mathbf{W}|\mathbf{x})$ are available where every $\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x}, \mathbf{y}')$ has full column rank for $\mathbf{y}' \in \mathfrak{X}_{\mathbf{Y}}$; (3) $Q_{\mathbf{z}'}(\mathbf{W})$ is available for $\mathbf{Z} \subseteq \mathbf{X} \cup \mathbf{Y}$ and $\mathbf{Z}' = (\mathbf{X} \cup \mathbf{Y})\backslash\mathbf{Z}$ with $\mathbf{z}'$ consistent with $\mathbf{x} \cup \mathbf{y}$ such that $(\mathbf{Z} \perp\!\!\!\perp \mathbf{W} \mid \mathbf{U}, \mathbf{Z}')$ in $\mathcal{H}$; and $\mathbf{U}$ is an adjustment admissible set for $Q_{\mathbf{z}'}(\mathbf{W})$ in $\mathcal{H}$.*

Every proxy matrix in the criterion has the same $\mathbf{U}$ column domain; full column rank is not assessed after dropping different $\mathbf{U}$ values in different slices. All elementwise divisions in this criterion are evaluated on $\operatorname{supp} Q(\mathbf{U})$; zero-weight terms are omitted from the final adjustment sum. The overlap condition in (1) ensures that the required denominators are positive.

The theorem presents a sound condition to elicit the causal effect through combining various distributions especially when the given situation is more restrictive.

Double-Proxy Setting We now generalize MGT criterion to utilize distributions other than the originally considered observational study. MGT criterion for a double-proxy setting relies on the following conditions to identify $P_x(y)$ with $P(X, Y, Z, W)$:

(C1) U is an adjustment set for $P_x(y)$ in $\mathcal{G}$;

(C2) $Y \perp\!\!\!\perp Z \mid U, X$ in $\mathcal{G}$;

(C3) $Z, X \perp\!\!\!\perp W \mid U$ in $\mathcal{G}$; and

(C4) $\mathbf{P}(W|Z, x)$ has rank $|\mathfrak{X}_U|$.

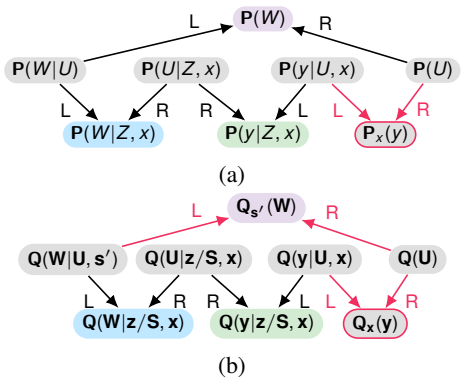


Figure 5: Schematics of (a) MGT criterion and (b) its generalization (simplified)

Under these conditions, algebraic relationships between a causal effect $P_x(y)$ and other distributions can be illustrated as in Fig. 5a where the distributions form a closed loop alternating between (i) distributions with an unmeasured confounder and (ii) given distributions and a query. Among the four multiplications, $P_x(y)$ corresponds to an adjustment criterion (C1) so as to $P_x(y) = \mathbf{P}(y|U, x)\mathbf{P}(U)$. Others are due to the chain-rule combined with CI, e.g., (C2) $\mathbf{P}(y|U, x) = \mathbf{P}(y|U, x, z)$ and (C3) $\mathbf{P}(W|U) = \mathbf{P}(W|U, x, z)$. With (C4), which implies that both $\mathbf{P}(W|U)$ and $\mathbf{P}(U|Z, x)$ are invertible under coarsening of W and Z if necessary [14, 1], the causal effect $P_x(y)$ can be expressed as Eq. (1) by subsequently rewriting $\mathbf{P}(U|Z, x)$ and $\mathbf{P}(y|U, x)$, and contracting $\mathbf{P}(W) = \mathbf{P}(W|U)\mathbf{P}(U)$ (see Sec. C for an illustration).

To examine the possible extension of the setting, we relax (C3), where the CI grants the use of matrix multiplication leading to a chain-rule $\mathbf{P}(W) = \mathbf{P}(W|U)\mathbf{P}(U)$ and $\mathbf{P}(W|Z, x) = \mathbf{P}(W|U)\mathbf{P}(U|Z, x)$. One can consider three relaxed versions of (C3), namely, $(Z \perp\!\!\!\perp W \mid U, X)$, $(X \perp\!\!\!\perp W \mid U, Z)$, and, ultimately, dropping (C3) illustrated respectively in Fig. 3c, 3d and 3e. For concreteness, consider the first relaxation $(Z \perp\!\!\!\perp W \mid U, X)$. The assumption grants $\mathbf{P}(W|U, Z, x)$=$\mathbf{P}(W|U, x)$. Given that a surrogate experiment $P_x(W)$ is accessible, and it can be decomposed as $\mathbf{P}_x(W) = \mathbf{P}(W|U, x)\mathbf{P}(U)$ where U is also an adjustment admissible set for $P_x(w)$, then, $P_x(y)$ is identified. Other two relaxations turned out to be much more challenging, yet the causal effect can be identified by exploiting different surrogate experiments or additional external

study (see Sec. C for detail). Motivated by these examples, we present a theorem that extends MGT criterion with varying degrees of the assumption and data collection.

Theorem 2 (Generalized MGT Criterion)**.** *Given a causal graph $\mathcal{G}$, let $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}, \mathbf{U}, \mathbf{R} \subset \mathbf{V}$ be disjoint sets of variables where $\mathbf{R}$ can be empty. Let $Q = P_{\mathbf{r}}$ for some $\mathbf{r} \in \mathfrak{X}_{\mathbf{R}}$ and $\mathcal{H} = \mathcal{G}\backslash\mathbf{R}$. Let $\mathbf{S} \subseteq \mathbf{X} \cup \mathbf{Z}$. A causal effect $Q_{\mathbf{x}}(\mathbf{y}) = P_{\mathbf{x},\mathbf{r}}(\mathbf{y})$ is identifiable in $\mathcal{G}$ if, for some $\mathbf{z}$, (1) $\mathbf{U}$ is an adjustment set for $Q_{\mathbf{x}}(\mathbf{y})$ in $\mathcal{H}$; (2) $(\mathbf{Y} \perp\!\!\!\perp \mathbf{Z} \mid \mathbf{U}, \mathbf{X})_{\mathcal{H}}$; (3) $(\mathbf{W} \perp\!\!\!\perp \mathbf{S} \mid \mathbf{U}, \mathbf{S}')_{\mathcal{H}}$ where $\mathbf{S}' = (\mathbf{X} \cup \mathbf{Z})\backslash\mathbf{S}$; (4) $\mathbf{U}$ is an adjustment set for $Q_{\mathbf{s}'}(\mathbf{W})$ in $\mathcal{H}$; (5) $\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{s}')$ is invertible; (6) $\mathbf{Q}(\mathbf{U}|\mathbf{z}/(\mathbf{S} \cup \mathbf{Z}'), \mathbf{x})$ is invertible for some $\mathbf{Z}' \subseteq \mathbf{Z}\backslash\mathbf{S}$ and $Q(\mathbf{y}|\mathbf{z}/(\mathbf{S} \cup \mathbf{Z}'), \mathbf{x})$, $Q(\mathbf{W}|\mathbf{z}/(\mathbf{S} \cup \mathbf{Z}'), \mathbf{x})$, and $Q_{\mathbf{s}'}(\mathbf{W})$ are available. Additionally, if $\mathbf{Z}' \neq \emptyset$, $Q(\mathbf{W}|\mathbf{U}, \mathbf{s}'/\mathbf{Z}')$ is available and every matrix slice indexed by $\mathbf{z}' \in \mathfrak{X}_{\mathbf{Z}'}$ is invertible; the $\mathbf{Z}'$ component of $\mathbf{s}'$ varies with that slice. Here $\mathbf{s}'$ is the assignment to $\mathbf{S}'$ induced by the selected $(\mathbf{x}, \mathbf{z})$. Within the slice enumeration, its $\mathbf{Z}'$ component is replaced by the enumerated value while all other components remain fixed.*

The theorem broadens the applicability of the MGT criterion as an identification template to a range of collection of distributions. Although the conditions involved in generalizing the MGT criterion are non-trivial, we can similarly draw a (simplified) big picture of the generalized criterion as in Fig. 5b. The key idea is relaxing the C3 condition for a set of variables while considering sets of variables (explicitly) and interventional distributions.

We have established generalized identification criteria exploiting proxy variables.[4] The purpose of both criteria aligns well with the philosophy of general identification [13, 12] so that they can be smoothly integrated into graphical approaches.

5 Pseudoinverse and Intermediary Criteria

Now, we present a novel identification condition for a challenging setting that neither previous matrical nor graphical approaches could handle. In the setting, the probability of interest is expressed as the multiplication of *three* matrices and is identifiable through employing the *pseudoinverse* of a matrix, *without* an invertibility assumption. Let $\mathbf{P}(\cdot)^{\dagger}$ denote the pseudoinverse of $\mathbf{P}(\cdot)$.[5]

Lemma 3 (Base Intermediary Criterion)**.** *Let $\{P_1, P_2, P_3, P_4\}$ be distributions. Let $\{\mathbf{P}_i\}_{i=1}^4$ be their matrix representations. If submatrices $\{\mathbf{P}'_i\}_{i=1}^4$ of $\{\mathbf{P}_i\}_{i=1}^4$ satisfy $\mathbf{P}'_1 = \mathbf{P}'_2\mathbf{P}'_3\mathbf{P}'_4$ and $\mathbf{P}'_2\mathbf{P}'_3$, $\mathbf{P}'_3\mathbf{P}'_4$, and $\mathbf{P}'_3$ are given, then, $\mathbf{P}'_1 = (\mathbf{P}'_2\mathbf{P}'_3)\mathbf{P}'_3{}^{\dagger}(\mathbf{P}'_3\mathbf{P}'_4)$.*

Proof. By the given condition, the associativity of matrix multiplications, and the property of pseudoinverse $\mathbf{P}\mathbf{P}^{\dagger}\mathbf{P} = \mathbf{P}$, $\mathbf{P}'_1 = \mathbf{P}'_2\mathbf{P}'_3\mathbf{P}'_4 = \mathbf{P}'_2(\mathbf{P}'_3\mathbf{P}'_3{}^{\dagger}\mathbf{P}'_3)\mathbf{P}'_4 = (\mathbf{P}'_2\mathbf{P}'_3)\mathbf{P}'_3{}^{\dagger}(\mathbf{P}'_3\mathbf{P}'_4)$. □

Although the lemma itself is rather general, we concretely characterize distributions satisfying Lemma 3 with respect to chain-rule (Sec. 5.1) and c-factorization (Sec. 5.2).

5.1 Chain-Rule-based Intermediary Criterion

We start by characterizing an intermediary criterion with a chain-rule using a simple illustrative example. Let Q be an arbitrary interventional distribution. One way to decompose $Q(a, b, c|d)$ into three probabilities is $Q(a|b, c, d)Q(b|c, d)Q(c|d)$. Let a probability of interest be its marginal $Q(a|d) = \sum_{b,c} Q(a, b, c|d)$ where the following distributions are available: $Q(B|C, d)$, $Q(A|C, d)$, and $Q(B|d)$. If the first term $Q(a|b, c, d)$ is equal to $Q(a|b, d)$, the term can be multiplied by $Q(b|d) = \sum_c Q(b|c, d)Q(c|d)$. Hence, the matricized

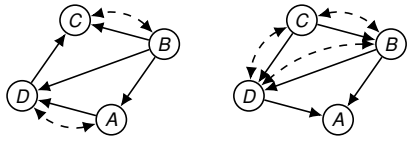


Figure 6: Causal diagrams where chain-rule intermediary criterion is applicable to identify $P(a|d)$ given $\mathbf{P}(a|C, d)$, $\mathbf{P}(B|C, d)$, and $\mathbf{P}(B|d)$.

[4] As mentioned earlier, [25] extends the MGT criterion by allowing observed confounders C. Roughly, this can be understood as replacing U in Fig. 5a with U and C where C is not necessarily marginalized out so that the resulting distribution may contain C.

[5] In case of continuous variables, $\mathbf{P}(B|A)$ can be understood as a linear operator on Hilbert spaces. See [15] for more detailed discussion on the existence and uniqueness of pseudoinverse for a bounded linear operator.

expression becomes,

$$\mathbf{Q}(A|d) = \overbrace{\mathbf{Q}(A|B,d)\underbrace{\mathbf{Q}(B|C,d)}_{\mathbf{Q}(B|d)}\mathbf{Q}(C|d)}^{\mathbf{Q}(A|C,d)} = \mathbf{Q}(A|C,d)\mathbf{Q}(B|C,d)^{\dagger}\mathbf{Q}(B|d).$$

for any $d \in \mathfrak{X}_D$. Two illustrative examples where this expression is applicable (i.e., $(C \perp\!\!\!\perp A \mid B, D)$ in $\mathcal{G}$) are shown in Fig. 6. Now, we propose a chain-rule intermediary criterion.

Lemma 4 (Chain-Rule Intermediary Criterion)**.** *Given a causal diagram $\mathcal{G}$, let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$, and $\mathbf{R}$ be disjoint subsets of $\mathbf{V}$ with $\mathbf{D}$ and $\mathbf{R}$ can be empty. Let $\mathbf{B} = \mathbf{B}' \cup \mathbf{B}''$ and $\mathbf{C} = \mathbf{C}' \cup \mathbf{C}''$ where $\mathbf{B}'$ and $\mathbf{C}'$ are not empty. Given an interventional distribution $Q = P_{\mathbf{r}}$, if $(\mathbf{C}' \perp\!\!\!\perp \mathbf{A} \mid \mathbf{B}, \mathbf{C}'', \mathbf{D})$ in $\mathcal{G}\backslash\mathbf{R}$ and $Q(\mathbf{a}, \mathbf{b}'', \mathbf{c}''|\mathbf{d}) = \sum_{\mathbf{b}',\mathbf{c}'} Q(\mathbf{a}|\mathbf{b},\mathbf{c},\mathbf{d})Q(\mathbf{b}|\mathbf{c},\mathbf{d})Q(\mathbf{c}|\mathbf{d})$, then,*

$$\mathbf{Q}(\mathbf{A}, \mathbf{b}'', \mathbf{c}''|\mathbf{d}) = \mathbf{Q}(\mathbf{A}, \mathbf{b}''|\mathbf{C}', \mathbf{c}'', \mathbf{d}) \cdot \mathbf{Q}(\mathbf{B}', \mathbf{b}''|\mathbf{C}', \mathbf{c}'', \mathbf{d})^{\dagger} \cdot \mathbf{Q}(\mathbf{B}', \mathbf{b}'', \mathbf{c}''|\mathbf{d}).$$

Additional CI allows the criterion to return a matrix, which can then be nested into other equation.

Corollary 1. *Given Lemma 4, if $(\mathbf{D} \perp\!\!\!\perp \mathbf{A}, \mathbf{B}'' \mid \mathbf{C})$ and $(\mathbf{D} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C})$ in $\mathcal{G}\backslash\mathbf{R}$, then*

$$\mathbf{Q}(\mathbf{A}, \mathbf{b}'', \mathbf{c}''|\mathbf{D}) = \mathbf{Q}(\mathbf{A}, \mathbf{b}''|\mathbf{C}', \mathbf{c}'') \cdot \mathbf{Q}(\mathbf{B}', \mathbf{b}''|\mathbf{C}', \mathbf{c}'')^{\dagger} \cdot \mathbf{Q}(\mathbf{B}', \mathbf{b}'', \mathbf{c}''|\mathbf{D}).$$

These results advance current knowledge of the causal or non-causal identification with fragments of information presented as conditional distributions, which is often under-studied.

5.2 C-Factorization-based Intermediary Criterion

We proceed to characterize an intermediary criterion where a matrix equation corresponds to a c-factorization resulting in three factors, extending the case of a pair of factors (Lemma 2). Concretely speaking, we are interested in the matrix form of

$$P_{\mathbf{x}_\ell}(\mathbf{y}_\ell) = \sum_{\mathbf{z},\mathbf{w}} P_{\mathbf{x}_i}(\mathbf{y}_i)P_{\mathbf{x}_j}(\mathbf{y}_j)P_{\mathbf{x}_k}(\mathbf{y}_k) = \sum_{\mathbf{z}} P_{\mathbf{x}_i}(\mathbf{y}_i) \sum_{\mathbf{w}} P_{\mathbf{x}_j}(\mathbf{y}_j)P_{\mathbf{x}_k}(\mathbf{y}_k), \tag{4}$$

where $\mathbf{Z} \subseteq \mathbf{X}_i \cap \mathbf{Y}_j$ and $\mathbf{W} \subseteq (\mathbf{X}_j \cap \mathbf{Y}_k)\backslash\mathbf{X}_i$ are disjoint sets of variables to be marginalized (other factor orders require the corresponding renamed set restrictions). In addition, available distributions are of the form $P_{\mathbf{X}_j}(\mathbf{Y}_j)$, $P_{\mathbf{X}_{ij}\backslash\mathbf{Y}_{ij}}(\mathbf{Y}_{ij}\backslash\mathbf{Z})$, and $P_{\mathbf{X}_{jk}\backslash\mathbf{Y}_{jk}}(\mathbf{Y}_{jk}\backslash\mathbf{W})$.

Theorem 3 (C-Factorization Intermediary Criterion)**.** *Let $\mathcal{G}$ be a causal diagram and $Q = P_{\mathbf{r}}$. Let $Q_{\mathbf{x}_\ell}(\mathbf{y}_\ell)$ be c-factorized as Eq.* (4)*. Let $\mathbf{X}_k^+$ be a subset of $\mathbf{X}_k$ excluding the rest five sets, $\{\mathbf{Y}_i, \mathbf{Y}_j, \mathbf{Y}_k, \mathbf{X}_i, \mathbf{X}_j\}$. $\mathbf{Y}_i^+$ is similarly defined. If $\mathbf{Z} \subseteq (\mathbf{X}_i \cap \mathbf{Y}_j)\backslash\mathbf{X}_k$ and $\mathbf{W} \subseteq (\mathbf{X}_j \cap \mathbf{Y}_k)\backslash\mathbf{X}_i$, then $\mathbf{Q}_{\mathbf{x}_\ell/\mathbf{X}_k^+}(\mathbf{y}_\ell/\mathbf{Y}_i^+)$, a submatrix of $\mathbf{Q}_{\mathbf{X}_\ell}(\mathbf{Y}_\ell)$, becomes*

$$\mathbf{Q}_{\mathbf{x}_\ell/\mathbf{X}_k^+}(\mathbf{y}_\ell/\mathbf{Y}_i^+) = \mathbf{Q}_{(\mathbf{x}_{ij}\backslash\mathbf{y}_{ij})/\mathbf{W}}((\mathbf{y}_{ij}\backslash\mathbf{Z})/\mathbf{Y}_i^+) \cdot \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z})^{\dagger} \cdot \mathbf{Q}_{(\mathbf{x}_{jk}\backslash\mathbf{y}_{jk})/\mathbf{X}_k^+}((\mathbf{y}_{jk}\backslash\mathbf{W})/\mathbf{Z}).$$

For the displayed factor order, both restrictions are needed: $\mathbf{W} \cap \mathbf{X}_i = \emptyset$ allows the first factor to be taken outside the inner sum over $\mathbf{w}$, and $\mathbf{Z} \cap \mathbf{X}_k = \emptyset$ allows the third factor to be taken outside the sum over $\mathbf{z}$. Other factor orders require the corresponding renamed restrictions. The set relationships are illustrated in Fig. 11 in the appendix.

The implication of this result is immediate. It was previously unknown whether the identification procedures by Lee and Bareinboim [12] (GID-PO) and Lee and Shpitser [11] (Lemma 3 and 6) taking marginal and interventional are complete. Now, we show a negative result.

Proposition 1. *GID-PO [12] and Lemma 3 and 6 [11] are not complete.*

Proof. Consider $X \to A \to B \to Y$ with inputs $P(X, B)$, $P(A, B)$, and $P(A, Y)$. Write $M = \mathbf{P}(B|A)$, $L = \mathbf{P}(y|B)$, and $N = \mathbf{P}(A|x)$. The available terms are $LM = \mathbf{P}(y|A)$, M, and $MN = \mathbf{P}(B|x)$. Therefore $\mathbf{P}(y|A)M^{\dagger}\mathbf{P}(B|x) = LMM^{\dagger}MN = LMN = P_x(y)$, without a full-rank requirement.

For the graphical procedures in [12, 11], the candidate outcome sets are $\{Y\}$, $\{A, Y\}$, $\{B, Y\}$, and $\{A, B, Y\}$. These require, respectively, the unavailable kernels $P_X(Y)$, $P_X(A)$, $P_B(Y)$, or both $P_X(A)$ and $P_B(Y)$. None is recoverable from one of the given pairwise inputs. With these unconditional inputs, the chain-rule closure of Definition 6 in Lee and Shpitser [11] is vacuous and eID reduces to mID. Thus neither procedure returns a formula on this instance. □

In this section, we developed novel intermediary criteria by characterizing both chain-rule and c-factorization with respect to matrix multiplications of three matrices exploiting the pseudoinverse, which has never been employed in the context of causal identification to the best of our knowledge.

6 A Unifying Causal Identification Algorithm

We present a causal identification algorithm ID-ME (Alg. 1) taking a collection of marginal, conditional, and interventional distributions $\mathbb{D}$ and causal graph $\mathcal{G}$.[6] The algorithm integrates different approaches such as generalized proxy-variable based criteria (Sec. 4 and [8, 10, 14]), intermediary criteria (Sec. 5), and factorization approaches [21, 13, 12, 11].

Taking a causal query $P_{\mathbf{x}}(\mathbf{y})$, causal graph $\mathcal{G}$, and distributions $\mathbb{D}$, it refines the given query by removing redundant interventions based on Rule 3 of do-calculus, i.e., $P_{\mathbf{x}}(\mathbf{y}) = P_{\mathbf{x}'}(\mathbf{y})$ for a minimal subset $\mathbf{X}' \subseteq \mathbf{X}$. Further, $\mathcal{G}'$ a copy of causal graph is obtained through pruning $\mathcal{G}$ only to the ancestors of $\mathbf{Y}$. However, note that the original $\mathcal{G}$ is kept intact to be used in the proxy criteria later where proxies can be non-ancestors of $\mathbf{Y}$, e.g., Fig. 1b. Then, the algorithm proceeds the following three parts: 1) expanding the given distributions $\mathbb{D}$; 2) c-factorizing the query $P_{\mathbf{x}}(\mathbf{y})$ and identifying each factor; and 3) combining identified c-factors $\mathbb{F}'$ to elicit the causal query.

First, the algorithm *expands* the given distributions (expand in Line 3). A given conditional distribution $P_{\mathbf{Z}}(\mathbf{V}'|\mathbf{T})$ can be understood as a distribution $P_{\mathbf{Z}}(\mathbf{V}')$ under a selection bias, which might hinder identification of a c-factor. By multiplying another distribution equal to, e.g., $P_{\mathbf{Z}}(\mathbf{T}''|\mathbf{T}')$, we can obtain a distribution with more outcome and less bias, $P_{\mathbf{Z}}(\mathbf{V}', \mathbf{T}''|\mathbf{T}')$. Chain rule closure [11] describes a state of a set of distributions where no new distribution with a smaller condition can be added to the collection. Function expand contains a procedure for chain rule closure along with chain-rule matrix inversion (Lemma 1) and chain-rule intermediary criterion (Lemma 4), enjoying the matrical approach especially relevant to conditional distributions.

The next part enumerates c-factors (Line 4) and attempts to identify every c-factor based on available distributions (Lines 5–8). An algorithm for identifying a c-factor given a *single* unconditional and non-marginal distribution has been thoroughly studied (e.g., Identify [27] and ID [21]), and is a building block for general identification [2, 13]. Dealing with a marginal or conditional distribution, one should consider (i) whether the condition in the conditional distribution in $\mathbb{D}$ can be negligible with respect to identifying the c-factor or (ii) whether matrix equations can be utilized (e.g., simple inverse or generalized proxy criteria). ID-RC (Line 7) is an identification module modified to handle a conditional distribution [4, 11], and PROXY (Line 8) applies Thms 1 and 2, including their rank, support, and per-slice requirements, and the double-proxy rank form of Eq. (1) under (C1)–(C4). ID-RC uses the conditional single-input c-factor identification procedure of Lemma 5 in Lee and Shpitser [11]; it is applied on the original graph $\mathcal{G}$, since the input distribution may contain non-ancestors of $\mathbf{Y}$.

The last part (Lines 9–10) attempts to map the identified c-factors to the causal effect $P_{\mathbf{x}}(\mathbf{y})$. Once a subset of c-factors are identified from the previous step, the algorithm proceeds to examine whether some of the unidentified factors in $\mathbb{F}''$ can be further inferred from the individually identified factors $\mathbb{F}'$, i.e., whether a simple inverse (CF-INV implementing Lemma 2) or the intermediary criterion over c-factors (CF-INT implementing Thm. 3) can be invoked (Line 9). Every CF-INT application must satisfy both $\mathbf{Z} \cap \mathbf{X}_k = \emptyset$ and $\mathbf{W} \cap \mathbf{X}_i = \emptyset$ for its chosen factor ordering. Then, we finally elicit a causal effect if a valid combination of identified c-factors exists (exact-cover in Line 10 checking Eq. (2)).

Every module is applied only when the graphical, rank, availability, and support conditions of its cited criterion hold. Successful factors retain their c-factorization and assignment information. The final exact cover must satisfy Eq. (2) in one eligible projection.

The algorithm coherently integrates existing methods and the newly developed machinery in the previous sections. The comparison concerns the cited criteria and algorithms in the discrete-variable setting considered here; subsumption refers to their successful identification cases. Whenever ID-ME returns a formula, evaluation of the formula leads to the quantity $P_{\mathbf{x}}(\mathbf{y})$. Further, ID-ME *strictly*

[6] We omit some details on whether some variables are fixed (e.g., $P_a(B)$ versus $P_A(B)$). Without loss of generality, every distribution contains no redundant conditions and interventions, which can be obtained through repeatedly applying rules of do-calculus.

Algorithm 1 ID-ME

function ID-ME($\mathbf{x}$, $\mathbf{y}$, $\mathcal{G}$, $\mathbb{D}$)
Input: $\mathbf{x}$, $\mathbf{y}$ value assignments for a query $P_{\mathbf{x}}(\mathbf{y})$; $\mathcal{G}$ a causal diagram; $\mathbb{D}$ a collection of distributions
Output: a formula for $P_{\mathbf{x}}(\mathbf{y})$ made with $\mathbb{D}$ or FAIL.
$\mathbf{X}, \mathcal{G}' \leftarrow \mathrm{an}^{\mathcal{G}_{\overline{\mathbf{x}}}}(\mathbf{Y}) \cap \mathbf{X}, \mathcal{G}[\mathrm{An}^{\mathcal{G}}(\mathbf{Y})]$
$\mathbb{D} \leftarrow$expand($\mathcal{G}, \mathbb{D}$) **unless** $\mathbb{D}$ is unconditional
$\mathbb{F}^{+} \leftarrow \mathbb{F}_{\mathcal{G}'}$ **if** $\mathbb{D}$ is unconditional and nonmarginal **else** $\bigcup\{\mathbb{F}_{\mathcal{G}'\langle \mathbf{V}' \rangle} \mid \mathbf{X} \cup \mathbf{Y} \subseteq \mathbf{V}' \subseteq \mathrm{An}^{\mathcal{G}}(\mathbf{Y})\}$
$\mathbb{F}', \mathbb{F}'' \leftarrow$ empty dictionary, $\mathbb{F}^{+}$
for all $P_{\mathbf{X}_i}(\mathbf{Y}_i) \in \mathbb{F}''$ and $\{P_{\mathbf{Z}}(\mathbf{V}'|\mathbf{T}) \in \mathbb{D} \mid \mathbf{Y}_i \subseteq \mathbf{V}', \mathbf{X}_i \subseteq \mathbf{Z} \cup \mathbf{V}' \cup \mathbf{T}\}$ **do**
    update $\mathbb{F}', \mathbb{F}''$ with ID-RC($P_{\mathbf{X}_i}(\mathbf{Y}_i)$, $\mathcal{G}$, $P_{\mathbf{Z}}(\mathbf{V}'|\mathbf{T})$)
for all $P_{\mathbf{X}_i}(\mathbf{Y}_i) \in \mathbb{F}''$ **do** update $\mathbb{F}', \mathbb{F}''$ with PROXY($P_{\mathbf{X}_i}(\mathbf{Y}_i)$, $\mathcal{G}$, $\mathbb{D} \cup \mathbb{F}'$)
repeat update $\mathbb{F}', \mathbb{F}''$ with CF-INT($\mathcal{G}', \mathbb{F}', \mathbb{F}''$) and CF-INV($\mathcal{G}', \mathbb{F}', \mathbb{F}''$) **until** $\mathbb{F}'$ not changed
return exact-cover($\mathcal{G}', P_{\mathbf{x}}(\mathbf{y}), \mathbb{F}'$)

subsumes the union of aforementioned methods in a way that not only returns it a formula whenever some of the methods returns one given the same problem instance but also it can output novel formulae for the problems that cannot be answered by any of them as demonstrated through examples and by Prop. 1.

Theorem 4. *ID-ME is sound.*

Theorem 5. *ID-ME strictly subsumes proxy criteria [8, 14], GID(-PO) [13, 12], mID, or eID [11].*

Finally, we discuss the time complexity of ID-ME through examining the complexity of algorithms concerning a subset of problem instances ID-ME can handle. First, the state of the art identification conditions with distributions marginal, experimental, but unconditional [12, 11] involve finding a latent projection where the corresponding set of c-factors are all identified. It is conjectured that it requires time exponential in $|\mathbf{V}|$ [12]. Next, dealing with conditional distributions through expanding them takes time exponential in $|\mathbf{V}|$ (e.g., chain rule closure [11]). As a result the algorithm generally runs in time exponential in $|\mathbf{V}|$, which we speculate that this complexity reflects a fundamental trade off between expressive power and tractability within causal inference, as already observed in [12, 11].

7 Conclusion

In this paper, we studied the use of matrix equations in causal identification given general distributions. In particular, we characterized matrix equations made of distributions driven by graphical constraints, deepening our understanding on algebraic constraints imposed by the graph (Lemmas 1 and 2 in Sec. 3). We generalized proxy-based identification methods to cover a broad spectrum of problem instances beyond the identification problems with an observation study and optional external study (Thms 1 and 2 in Sec. 4). We developed novel intermediary criteria that identify a query by utilizing the pseudoinverse of the center matrix within the multiplication of three matrices (Lemma 4 and Thm. 3 in Sec. 5). Finally, we provide a causal identification algorithm (Alg. 1 in Sec. 6) integrating several existing identification results to produce an identification formula for a causal query given a causal graph and distributions that can be marginal, experimental, and conditional harnessing the power of matrix equations and (pseudo)inverses. As a future research direction, it will be crucial to devise a statistically efficient estimator for identification formulae involving matrix inversion.

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Appendix for "Causal Identification with Matrix Equations"

A Preliminaries

We formally introduce essential definitions like d-separation, do-calculus, latent projection, and adjustment criterion.

d-Separation We begin with the definition of d-separation [16].

Definition 1 (d-separation)**.** Two sets of vertices $\mathbf{X}, \mathbf{Y}$ are said to be d-separated by another set $\mathbf{Z}$ in a directed acyclic graph $\mathcal{G}$, denoted by $(\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z})_{\mathcal{G}}$, if every path $\mathbf{P}$ from vertices in $\mathbf{X}$ to vertices in $\mathbf{Y}$ are blocked where blockage occurs when one of the following holds:

1. $\mathbf{P}$ contains at least one arrow-emitting node that is in $\mathbf{Z}$, or
2. $\mathbf{P}$ contains at least one collider that is outside $\mathbf{Z}$ and has no descendant in $\mathbf{Z}$.

For causal diagrams with bidirected edges, we use the same separation notation on a latent-DAG expansion, replacing each bidirected edge by an unobserved common parent. This is the corresponding m-separation criterion on the observed graph.

Do-calculus Do-calculus [17] is an essential machinery to reason about the equivalence of conditional interventional probabilities induced by any model conforming to a given causal diagram. Do-calculus consists of three rules where each rule dictates an equality between two probability distributions under a certain graphical test. These rules are

R1 (Insertion/deletion of observations):

$$P_{\mathbf{x}}(\mathbf{y}|\mathbf{z}, \mathbf{w}) = P_{\mathbf{x}}(\mathbf{y}|\mathbf{w}) \qquad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X}, \mathbf{W} \text{ in } \mathcal{G}_{\overline{\mathbf{X}}}$$

R2 (Action/observation exchange):

$$P_{\mathbf{x}}(\mathbf{y}|\mathbf{z}, \mathbf{w}) = P_{\mathbf{x},\mathbf{z}}(\mathbf{y}|\mathbf{w}) \qquad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X}, \mathbf{W} \text{ in } \mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$$

R3 (Insertion/deletion of actions):

$$P_{\mathbf{x}}(\mathbf{y}|\mathbf{w}) = P_{\mathbf{x},\mathbf{z}}(\mathbf{y}|\mathbf{w}) \qquad \text{if } \mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X}, \mathbf{W} \text{ in } \mathcal{G}_{\overline{\mathbf{X}}\overline{\mathbf{Z(W)}}}$$

where $\mathbf{Z}(\mathbf{W})$ is a subset of $\mathbf{Z}$ that is not ancestor of $\mathbf{W}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$.

Latent Projection We formally define latent projection as below.

Definition 2 (Latent Projection (adopted from [29]))**.** The projection of a causal graph $\mathcal{G}$ over $\mathbf{V}$ on $\mathbf{V}' \subseteq \mathbf{V}$, denoted by $\mathcal{G}\langle\mathbf{V}'\rangle$, is a causal diagram over $\mathbf{V}'$ such that for every pair $X, Y \in \mathbf{V}'$:

1. There exists a directed edge $X \to Y$ in $\mathcal{G}\langle\mathbf{V}'\rangle$ if there exists a directed path from X to Y in $\mathcal{G}$ such that every vertex other than X and Y on the path is not in $\mathbf{V}'$.
2. There exists a bidirected edge $X \leftrightarrow Y$ in $\mathcal{G}\langle\mathbf{V}'\rangle$ if there exists a divergent path[7] between X and Y in $\mathcal{G}$ such that every vertex other than X and Y on the path is not in $\mathbf{V}'$.

Back-door and Adjustment Criterion We reproduce the back-door criterion.

Definition 3 (Back-door Criterion)**.** Let $\mathbf{X}$ and $\mathbf{Y}$ be disjoint sets of vertices in a graph $\mathcal{G}$. Then a set $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if

1. No element in $\mathbf{Z}$ is a descendant of $\mathbf{X}$.
2. $\mathbf{Z}$ d-separates all back-door paths from $\mathbf{X}$ to $\mathbf{Y}$.

The definition of adjustment criterion is from [22, 20].

[7] A path between X and Y is said to be *divergent* if it is made of two directed paths $(W_1, \ldots, X)$ and $(W_2, \ldots, Y)$ in $\mathcal{G}$ such that $W_1 = W_2$ or $W_1 \leftrightarrow W_2$.

Definition 4 (Proper Causal Path)**.** Let $\mathbf{X}$, $\mathbf{Y}$ be sets of vertices. A causal path from a vertex in $\mathbf{X}$ to a vertex in $\mathbf{Y}$ is called *proper* if it does not intersect $\mathbf{X}$ except at the end point.

Definition 5 (Adjustment Criterion)**.** $\mathbf{Z}$ satisfies the adjustment criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if

1. No element in $\mathbf{Z}$ is a descendant in $\mathcal{G}_{\overline{\mathbf{X}}}$ of any $W \notin \mathbf{X}$ which lies on a proper causal path from $\mathbf{X}$ to $\mathbf{Y}$.
2. All proper non-causal paths in $\mathcal{G}$ from $\mathbf{X}$ to $\mathbf{Y}$ are blocked by $\mathbf{Z}$.

B Characterization of Matrix Equations of Graphical Constraints

Lemma 1. *Given a causal diagram $\mathcal{G}$, let $Q = P_{\mathbf{r}}$ for some $\mathbf{r} \in \mathfrak{X}_{\mathbf{R}}$ where $\mathbf{R} \subsetneq \mathbf{V}$. Let $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}$ be disjoint subsets of $\mathbf{V} \backslash \mathbf{R}$. If $(\mathbf{D} \perp\!\!\!\perp \mathbf{A} \mid \mathbf{B}, \mathbf{C}, \mathbf{E})$ and $(\mathbf{E} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}, \mathbf{D})$ in $\mathcal{G} \backslash \mathbf{R}$, then, $\mathbf{Q}(\mathbf{A}, \mathbf{b}' | \mathbf{c}, \mathbf{D}, \mathbf{e}) = \mathbf{Q}(\mathbf{A} | \mathbf{b}', \mathbf{B}'', \mathbf{c}, \mathbf{e}) \mathbf{Q}(\mathbf{B}'', \mathbf{b}' | \mathbf{c}, \mathbf{D})$.*

Proof.

$$
\begin{aligned}
Q(\mathbf{a}, \mathbf{b} | \mathbf{c}, \mathbf{d}, \mathbf{e}) &= Q(\mathbf{a} | \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}) Q(\mathbf{b} | \mathbf{c}, \mathbf{d}, \mathbf{e}) && \text{Chain Rule} \\
Q(\mathbf{a}, \mathbf{b}' | \mathbf{c}, \mathbf{d}, \mathbf{e}) &= \sum_{\mathbf{b}''} Q(\mathbf{a} | \mathbf{b}', \mathbf{b}'', \mathbf{c}, \mathbf{d}, \mathbf{e}) Q(\mathbf{b}', \mathbf{b}'' | \mathbf{c}, \mathbf{d}, \mathbf{e}) && \text{Marginalization} \\
&= \sum_{\mathbf{b}''} Q(\mathbf{a} | \mathbf{b}', \mathbf{b}'', \mathbf{c}, \mathbf{e}) Q(\mathbf{b}', \mathbf{b}'' | \mathbf{c}, \mathbf{d}) && \text{Conditional Independence} \\
&= \mathbf{Q}(\mathbf{a} | \mathbf{b}', \mathbf{B}'', \mathbf{c}, \mathbf{e}) \mathbf{Q}(\mathbf{b}', \mathbf{B}'' | \mathbf{c}, \mathbf{d}) && \text{Matrix Representation} \\
\mathbf{Q}(\mathbf{A}, \mathbf{b}' | \mathbf{c}, \mathbf{D}, \mathbf{e}) &= \mathbf{Q}(\mathbf{A} | \mathbf{b}', \mathbf{B}'', \mathbf{c}, \mathbf{e}) \mathbf{Q}(\mathbf{b}', \mathbf{B}'' | \mathbf{c}, \mathbf{D}). && \text{Matricization}
\end{aligned}
$$

□

Lemma 2 (Matrix Equation of C-Factorization with Two Factors)**.** *Given a causal diagram $\mathcal{G}$ and an experimental distribution $Q = P_{\mathbf{r}}$ where a causal effect is c-factorized as $Q_{\mathbf{x}}(\mathbf{y}) = \sum_{\mathbf{z}} Q_{\mathbf{x}_i}(\mathbf{y}_i) Q_{\mathbf{x}_j}(\mathbf{y}_j)$ in a projection $\mathcal{H}$ of $\mathcal{G} \backslash \mathbf{R}$, the effect can be represented as a matrix multiplication, if $\mathbf{Z} \subseteq \mathbf{X}_i \cap \mathbf{Y}_j$. Further, the corresponding matrix equation is*

$$
\mathbf{Q}_{(\mathbf{x}_{ij} \backslash \mathbf{y}_{ij}) / (\mathbf{X}_j \backslash \mathbf{X}_i \backslash \mathbf{Y}_i)}((\mathbf{y}_{ij} \backslash \mathbf{Z}) / (\mathbf{Y}_i \backslash \mathbf{X}_j)) = \mathbf{Q}_{\mathbf{x}_i / \mathbf{Z}}(\mathbf{y}_i / (\mathbf{Y}_i \backslash \mathbf{X}_j)) \mathbf{Q}_{\mathbf{x}_j / (\mathbf{X}_j \backslash \mathbf{X}_i \backslash \mathbf{Y}_i)}(\mathbf{y}_j / \mathbf{Z}).
$$

Proof. Since the columns of the left matrix and the rows of the right matrix must be matched to $\mathbf{Z}$, the rest of sets $\mathbf{X}_i \backslash \mathbf{Z}$ and $\mathbf{Y}_j \backslash \mathbf{Z}$ are fixed. Further, to avoid the resulting matrix sharing the same variables, $\mathbf{Y}_i \cap \mathbf{X}_j$ is fixed. Then, using the colors as illustrated in Fig. 2a,

$$
\begin{aligned}
&\mathbf{Q}_{(\mathbf{x}_{ij} \backslash \mathbf{y}_{ij}) / (\mathbf{X}_j \backslash \mathbf{X}_i \backslash \mathbf{Y}_i)}((\mathbf{y}_{ij} \backslash \mathbf{Z}) / (\mathbf{Y}_i \backslash \mathbf{X}_j)) \\
&\quad = \mathbf{Q}_{\mathbf{x}_i \backslash \mathbf{Z}, \mathbf{Z}}(\mathbf{Y}_i \backslash \mathbf{X}_j, \mathbf{y}_i \cap \mathbf{x}_j) \mathbf{Q}_{\mathbf{X}_j \backslash \mathbf{X}_i \backslash \mathbf{Y}_i, \mathbf{x}_i \cap \mathbf{x}_j, \mathbf{y}_i \cap \mathbf{x}_j}(\mathbf{y}_j \backslash \mathbf{Z}, \mathbf{Z}) \\
&\quad = \mathbf{Q}_{\mathbf{x}_i / \mathbf{Z}}(\mathbf{y}_i / (\mathbf{Y}_i \backslash \mathbf{X}_j)) \mathbf{Q}_{\mathbf{x}_j / (\mathbf{X}_j \backslash \mathbf{X}_i \backslash \mathbf{Y}_i)}(\mathbf{y}_j / \mathbf{Z}).
\end{aligned}
$$

□

C Generalized Proxy-based Criteria

We explain the modified schematic of the single-proxy setting in Fig. 7. We reproduce Fig. 4 with a twist by replacing $\mathbf{P}(W|U)$ to $\mathbf{P}(W|U)^{-1}$ so as to easily observe how the first three unobserved distributions with U in their measurements are obtained. Then, $\mathbf{P}(y|U, x)$ is computed. Finally, $P_x(y)$ is identified.

C.1 Single-Proxy Setting

We derive two identification formulae when the assumption is relaxed in two different ways.

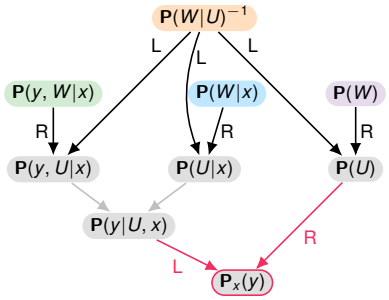
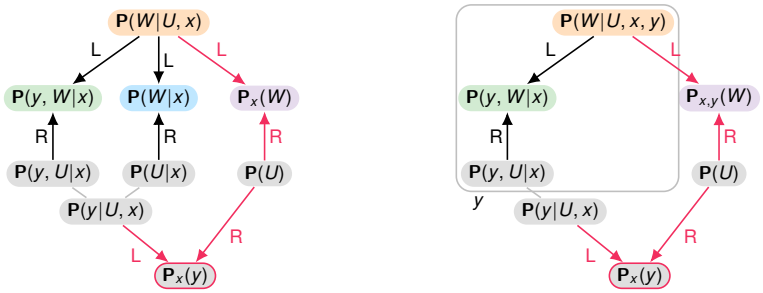


Figure 7: A schematic for identifiability with $P(W|U)^{-1}$, an external study $P(W|U)$ inverted. $\mathbf{P}(y|U,x)$ is based on the transpose of the elementwise quotient of $\mathbf{P}(y,U|x)$ and $\mathbf{P}(U|x)$.



(a) $(Y \perp\!\!\!\perp W \mid U, X)$ (b) without $(X, Y \perp\!\!\!\perp W \mid U)$

Figure 8: Schematics for the single-proxy setting with different CI statements, (a) for Fig. 3a and (b) for Fig. 3b. In (a), $\mathbf{P}(W|x)$ is obtained by marginalizing $\mathbf{P}(Y,W|x)$ over Y. In (b), the gray rectangle indicates enumeration over y to recover $\mathbf{P}(Y,U|x)$.

Case $(Y \perp\!\!\!\perp W \mid U, X)$**:** Consider a case $(Y \perp\!\!\!\perp W \mid U, X)$ as illustrated in Fig. 3a. Further, unlike the original setting, suppose an invertible matrix $\mathbf{P}(W|U,x)$ is available instead of $\mathbf{P}(W|U)$. Then, the identification cannot be completed since $\mathbf{P}(W) = \mathbf{P}(W|U)\mathbf{P}(U)$ becomes infeasible. Nevertheless, if U is an admissible set for a surrogate experiment $P_x(w)$ and an experimental distribution $P_x(W)$ is available, then $P_x(y)$ is identified as (fraction corresponds to elementwise division, $\oslash$)

$$\left(\frac{\mathbf{P}(W|U,x)^{-1}\mathbf{P}(y,W|x)}{\mathbf{P}(W|U,x)^{-1}\mathbf{P}(W|x)}\right)^{\top}\mathbf{P}(W|U,x)^{-1}\mathbf{P}_x(W).$$

Case without $(X, Y \perp\!\!\!\perp W \mid U)$**:** Now, suppose that neither X nor Y is conditionally independent to W given U. Each $\mathbf{P}(W|U,x,y)$ is assumed to have full column rank, and $\mathbf{P}(W|U,x,y)$ is required for obtaining $\mathbf{P}(y,U|x)$ through $\mathbf{P}(y,W|x) = \mathbf{P}(W|U,x,y)\mathbf{P}(y,U|x)$. Since $P(W|U,x,y)$ is no longer equal to $P(W|U,x)$, one may additionally require $\mathbf{P}(W|U,x)$ to identify $\mathbf{P}(U|x)$. By the way, having $P(W|U,x,Y)$ is sufficient to obtain $\mathbf{P}(U|x)$ through marginalizing out Y from $P(Y,U|x) = [P(y',U|x)]_{y'}$ so as to compute $P(y|U,x)$. Similar to the previous setting, the existence of a surrogate experiment $P_{x,y}(W)$ where U is an admissible set for $P_{x,y}(W)$ in $\mathcal{G}$ allows the identification of $\mathbf{P}(U)$, thus eliciting $P_x(y)$,

$$\left(\frac{\mathbf{P}(W|U,x,y)^{\dagger}\mathbf{P}(y,W|x)}{\sum_{y'}\mathbf{P}(W|U,x,y')^{\dagger}\mathbf{P}(y',W|x)}\right)^{\top}\mathbf{P}(W|U,x,y)^{\dagger}\mathbf{P}_{x,y}(W).$$

We depict two relaxations of a single-proxy setting in Fig. 8. Fig. 8a replaces $\mathbf{P}(W)$ with $\mathbf{P}_x(W)$ based on the adjustment criterion. With fixed x, the relaxed condition does not affect how $\mathbf{P}(y,W|x)$ is computed. In Fig. 8b, non-redundant y in $\mathbf{P}(W|U,x,y)$ prohibits computing $\mathbf{P}(W|x)$. Even though we obtain $\mathbf{P}(W|x)$ from $\mathbf{P}(Y,W|x)$, the expression $\mathbf{P}(W|x) = \mathbf{P}(W|U,x,y)\mathbf{P}(U|x)$ is invalid. Once we can enumerate $\mathbf{P}(Y,U|x)$, we can recover $\mathbf{P}(y|U,x)$ so as to compute $P_x(y)$. Hence, $\mathbf{P}(W|U,x,Y)$ and $\mathbf{P}(Y,W|x)$ are sufficient.

Theorem 1. *Given $\mathcal{G}$, let $\mathbf{X}$, $\mathbf{Y}$, $\mathbf{W}$, $\mathbf{U}$, and $\mathbf{R}$ be disjoint subsets of $\mathbf{V}$, $Q = P_{\mathbf{r}}$ for some $\mathbf{r} \in \mathfrak{X}_{\mathbf{R}}$, and $\mathcal{H} = \mathcal{G}\backslash\mathbf{R}$. A causal effect $P_{\mathbf{r},\mathbf{x}}(\mathbf{y}) = Q_{\mathbf{x}}(\mathbf{y})$ is identifiable if (1) $\mathbf{U}$ is an adjustment admissible set for $Q_{\mathbf{x}}(\mathbf{y})$ in $\mathcal{H}$, and $Q(\mathbf{u}|\mathbf{x}) > 0$ for every $\mathbf{u}$ with $Q(\mathbf{u}) > 0$; (2a) $(\mathbf{Y} \perp\!\!\!\perp \mathbf{W} \mid \mathbf{U}, \mathbf{X})_{\mathcal{H}}$ and $Q(\mathbf{W}|\mathbf{U}, \mathbf{x})$, $Q(\mathbf{y}, \mathbf{W}|\mathbf{x})$ and $Q(\mathbf{W}|\mathbf{x})$ are available where $\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x})$ has full column rank; or (2b) $(\mathbf{Y} \not\perp\!\!\!\perp \mathbf{W} \mid \mathbf{U}, \mathbf{X})_{\mathcal{H}}$ and $Q(\mathbf{W}|\mathbf{U}, \mathbf{x}, \mathbf{Y})$ and $Q(\mathbf{Y}, \mathbf{W}|\mathbf{x})$ are available where every $\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x}, \mathbf{y}')$ has full column rank for $\mathbf{y}' \in \mathfrak{X}_{\mathbf{Y}}$; (3) $Q_{\mathbf{z}'}(\mathbf{W})$ is available for $\mathbf{Z} \subseteq \mathbf{X} \cup \mathbf{Y}$ and $\mathbf{Z}' = (\mathbf{X} \cup \mathbf{Y})\backslash\mathbf{Z}$ with $\mathbf{z}'$ consistent with $\mathbf{x} \cup \mathbf{y}$ such that $(\mathbf{Z} \perp\!\!\!\perp \mathbf{W} \mid \mathbf{U}, \mathbf{Z}')$ in $\mathcal{H}$; and $\mathbf{U}$ is an adjustment admissible set for $Q_{\mathbf{z}'}(\mathbf{W})$ in $\mathcal{H}$.*

Proof. Operator $\oslash$ denotes elementwise division on $\operatorname{supp} Q(\mathbf{U})$. The overlap condition in (1) makes every denominator on this support positive; terms with $Q(\mathbf{u}) = 0$ are omitted. Every pseudoinverse below is a left inverse because its matrix has full column rank. We begin with identifying $\mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x})$ using (2a) and (2b), separately. **Case (2a):** Given (2a), $\mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x})$ is obtainable. First, we can compute $\mathbf{Q}(\mathbf{y}, \mathbf{U}|\mathbf{x})$ as

$$\begin{aligned} \mathbf{Q}(\mathbf{y}, \mathbf{W}|\mathbf{x}) &= \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x})\mathbf{Q}(\mathbf{y}, \mathbf{U}|\mathbf{x}), && \text{Chain Rule and (2a)} \\ \mathbf{Q}(\mathbf{y}, \mathbf{U}|\mathbf{x}) &= \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x})^{\dagger}\mathbf{Q}(\mathbf{y}, \mathbf{W}|\mathbf{x}). && \text{(2a) Left inversion} \end{aligned} \tag{5}$$

Next, $\mathbf{Q}(\mathbf{U}|\mathbf{x})$,

$$\begin{aligned} \mathbf{Q}(\mathbf{W}|\mathbf{x}) &= \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x})\mathbf{Q}(\mathbf{U}|\mathbf{x}), && \text{Chain Rule} \\ \mathbf{Q}(\mathbf{U}|\mathbf{x}) &= \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x})^{\dagger}\mathbf{Q}(\mathbf{W}|\mathbf{x}). && \text{(2a) Left inversion} \end{aligned} \tag{6}$$

Combining Eq. (5) and Eq. (6), we can get $\mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x})$,

$$\mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x}) = (\mathbf{Q}(\mathbf{y}, \mathbf{U}|\mathbf{x}) \oslash \mathbf{Q}(\mathbf{U}|\mathbf{x}))^{\top}. \tag{7}$$

Case (2b): Similarly, given (2b),

$$\begin{aligned} \mathbf{Q}(\mathbf{y}', \mathbf{W}|\mathbf{x}) &= \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x}, \mathbf{y}')\mathbf{Q}(\mathbf{y}', \mathbf{U}|\mathbf{x}), && \text{Chain Rule} \\ \mathbf{Q}(\mathbf{Y}, \mathbf{U}|\mathbf{x}) &= [\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x}, \mathbf{y}')^{\dagger}\mathbf{Q}(\mathbf{y}', \mathbf{W}|\mathbf{x})]_{\mathbf{y}'}, && \text{(2b) Left inversion} \quad (8) \\ \mathbf{Q}(\mathbf{U}|\mathbf{x}) &\leftarrow \mathbf{Q}(\mathbf{Y}, \mathbf{U}|\mathbf{x}). && \text{Marginalization} \quad (9) \end{aligned}$$

Combining Eq. (8) instantiated at $\mathbf{y}$ and Eq. (9),

$$\mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x}) = (\mathbf{Q}(\mathbf{y}, \mathbf{U}|\mathbf{x}) \oslash \mathbf{Q}(\mathbf{U}|\mathbf{x}))^{\top}. \tag{10}$$

Now let $\mathbf{Z}' = (\mathbf{X} \cup \mathbf{Y})\backslash\mathbf{Z}$, with $\mathbf{z}'$ consistent with $\mathbf{x}, \mathbf{y}$. In case (2a), put $\mathbf{A} = \mathbf{X} \cap \mathbf{Z}$, $\mathbf{B} = \mathbf{X} \cap \mathbf{Z}'$, and $\mathbf{C} = \mathbf{Y} \cap \mathbf{Z}'$. Conditions (2a) and (3) give $(\mathbf{W} \perp\!\!\!\perp \mathbf{C} \mid \mathbf{U}, \mathbf{A}, \mathbf{B})_{\mathcal{H}}$ and $(\mathbf{W} \perp\!\!\!\perp \mathbf{A} \mid \mathbf{U}, \mathbf{B}, \mathbf{C})_{\mathcal{H}}$. Intersection of graphical separation yields $(\mathbf{W} \perp\!\!\!\perp \mathbf{A}, \mathbf{C} \mid \mathbf{U}, \mathbf{B})_{\mathcal{H}}$. Thus $\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{z}') = \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x})$ on the stipulated conditioning supports, without conditioning on the full outcome value $\mathbf{y}$. In case (2b), Condition (3) gives $\mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{z}') = \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{x}, \mathbf{y})$. Write K for the corresponding available full-column-rank proxy matrix. Then Condition (3) gives

$$\begin{aligned} \mathbf{Q}_{\mathbf{z}'}(\mathbf{W}) &= K\mathbf{Q}(\mathbf{U}), && \text{(3) Adjustment and CI} \\ \mathbf{Q}(\mathbf{U}) &= K^{\dagger}\mathbf{Q}_{\mathbf{z}'}(\mathbf{W}). && \text{Left inversion} \end{aligned} \tag{11}$$

Finally, we obtain the causal query based on Condition (1) by plugging in $\mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x})$ from Condition (2) (either Eq. (7) or Eq. (10)) and $\mathbf{Q}(\mathbf{U})$ (Eq. (11)) from Condition (3),

$$Q_{\mathbf{x}}(\mathbf{y}) = \mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x})\mathbf{Q}(\mathbf{U}). \qquad \text{(1) Adjustment} \tag{12}$$

□

C.2 Double-Proxy Setting

We provide how MGT criterion's derivation works as an illustration in Fig. 9. It is noteworthy to observe that the three distributions on the left side $\mathbf{P}(W|U)$, $\mathbf{P}(W|Z, x)$, and $\mathbf{P}(U|Z, x)$ are square matrices so that the double-inversion of $\mathbf{P}(W|U)$ becomes possible.

Motivating Examples We provide two of the three variants of MGT criterion with respect to the variations of assumption (C3).

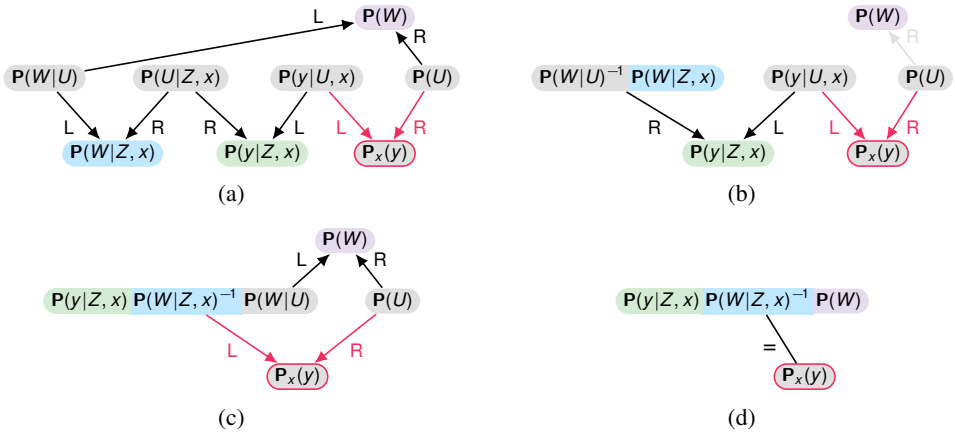


Figure 9: Schematics of the double-proxy setting (i.e., MGT criterion). Given the basic setting (a), substituting (b) $\mathbf{P}(U|Z,x)$ and (c) $\mathbf{P}(y|U,x)$, and (d) contracting $\mathbf{P}(W|U)\mathbf{P}(U)$ lead to the identification of $P_x(y)$.

Case $(Z \perp\!\!\!\perp W \mid U, X)$ The assumption grants $\mathbf{P}(W|U,Z,x)$=$\mathbf{P}(W|U,x)$. Given that a surrogate experiment $P_x(W)$ is accessible and it can be decomposed as $\mathbf{P}_x(W) = \mathbf{P}(W|U,x)\mathbf{P}(U)$ where U is also an admissible set for $P_x(w)$, then, $P_x(y)$ is identified. An example is illustrated in Fig. 3c where an additional directed edge from X to W is allowed.

Case $(X \not\perp\!\!\!\perp W \mid U, Z)$ One might surmise that we can similarly assume the existence of a surrogate experiment and its factorization based on an adjustment criterion,

$$\mathbf{P}_z(W) = \mathbf{P}(W|U,z)\mathbf{P}(U)$$

However, this dependence $(Z \not\perp\!\!\!\perp W \mid U)$ prohibits a matrix multiplication involving $\mathbf{P}(W|Z,x)$, i.e., Z needs to be instantiated as $\mathbf{P}(U|Z,x) = [\mathbf{P}(U|z',x)]_{z'}$ where $\mathbf{P}(U|z,x) = \mathbf{P}(W|U,z)^{-1}\mathbf{P}(W|z,x)$. Ultimately, we are unable to invoke the *double-inversion* trick as depicted in Fig. 9 to contract $\mathbf{P}(W|U,z)$ and $\mathbf{P}(U)$ into a U-free term.

This challenging situation can be addressed by assuming the existence of an external study $P(W|U,Z)$. Then,

$$P_x(y) = \mathbf{P}(y|Z,x)\mathbf{P}(U|Z,x)^{-1}\mathbf{P}(U) \tag{13}$$

where $\mathbf{P}(U|Z,x)$ is plugged-in and $\mathbf{P}(U) = \mathbf{P}(W|U,z')^{-1}\mathbf{P}_{z'}(W)$ for some $z' \in \mathfrak{X}_Z$. An example is shown in Fig. 3d. Note the absence of a directed edge from W to Y to preserve (C2).

Further dropping the condition $(X \perp\!\!\!\perp W \mid U, Z)$ yields a similar result where $\mathbf{P}(W|U,z,x)$ and $\mathbf{P}_{x,z}(W)$ replace $\mathbf{P}(W|U,z)$ and $\mathbf{P}_z(W)$, respectively, since x, unlike z, is *fixed* throughout the derivation (see Fig. 3e for an instance).

For the rank-form expression in Eq. (1), write $M = AB$ with $A = \mathbf{P}(W|U)$ and $B = \mathbf{P}(U|Z,x)$. The condition $\operatorname{rank} M = |\mathfrak{X}_U|$ makes A full column rank and B full row rank. Hence $M^\dagger = B^\dagger A^\dagger$, and $\mathbf{P}(y|Z,x)M^\dagger\mathbf{P}(W) = \mathbf{P}(y|U,x)BB^\dagger A^\dagger A\mathbf{P}(U) = P_x(y)$.

Theorem 2 (Generalized MGT Criterion)**.** *Given a causal graph $\mathcal{G}$, let $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}, \mathbf{U}, \mathbf{R} \subset \mathbf{V}$ be disjoint sets of variables where $\mathbf{R}$ can be empty. Let $Q = P_{\mathbf{r}}$ for some $\mathbf{r} \in \mathfrak{X}_{\mathbf{R}}$ and $\mathcal{H} = \mathcal{G}\backslash\mathbf{R}$. Let $\mathbf{S} \subseteq \mathbf{X}\cup\mathbf{Z}$. A causal effect $Q_{\mathbf{x}}(\mathbf{y}) = P_{\mathbf{x},\mathbf{r}}(\mathbf{y})$ is identifiable in $\mathcal{G}$ if, for some $\mathbf{z}$, (1) $\mathbf{U}$ is an adjustment set for $Q_{\mathbf{x}}(\mathbf{y})$ in $\mathcal{H}$; (2) $(\mathbf{Y} \perp\!\!\!\perp \mathbf{Z} \mid \mathbf{U}, \mathbf{X})_{\mathcal{H}}$; (3) $(\mathbf{W} \perp\!\!\!\perp \mathbf{S} \mid \mathbf{U}, \mathbf{S}')_{\mathcal{H}}$ where $\mathbf{S}' = (\mathbf{X}\cup\mathbf{Z})\backslash\mathbf{S}$; (4) $\mathbf{U}$ is an adjustment set for $Q_{\mathbf{s}'}(\mathbf{W})$ in $\mathcal{H}$; (5) $\mathbf{Q}(\mathbf{W}|\mathbf{U},\mathbf{s}')$ is invertible; (6) $\mathbf{Q}(\mathbf{U}|\mathbf{z}/(\mathbf{S}\cup\mathbf{Z}'),\mathbf{x})$ is invertible for some $\mathbf{Z}' \subseteq \mathbf{Z}\backslash\mathbf{S}$ and $Q(\mathbf{y}|\mathbf{z}/(\mathbf{S}\cup\mathbf{Z}'),\mathbf{x})$, $Q(\mathbf{W}|\mathbf{z}/(\mathbf{S}\cup\mathbf{Z}'),\mathbf{x})$, and $Q_{\mathbf{s}'}(\mathbf{W})$ are available. Additionally, if $\mathbf{Z}' \neq \emptyset$, $Q(\mathbf{W}|\mathbf{U},\mathbf{s}'/\mathbf{Z}')$ is available and every matrix slice indexed by $\mathbf{z}' \in \mathfrak{X}_{\mathbf{Z}'}$ is invertible; the $\mathbf{Z}'$ component of $\mathbf{s}'$ varies with that slice. Here $\mathbf{s}'$ is the assignment to $\mathbf{S}'$ induced by the selected $(\mathbf{x},\mathbf{z})$. Within the slice enumeration, its $\mathbf{Z}'$ component is replaced by the enumerated value while all other components remain fixed.*

Proof. Fix the selected value $\mathbf{z}_0$ in the theorem. Let $\mathbf{T} = \mathbf{Z} \cap \mathbf{S}$, $\mathbf{E} = \mathbf{Z}'$, and $\mathbf{F} = \mathbf{Z}\backslash(\mathbf{T}\cup\mathbf{E})$. Write $\mathbf{f} = \mathbf{z}_0\backslash(\mathbf{T}\cup\mathbf{E})$ and $\mathbf{e}_0 = \mathbf{z}_0\backslash(\mathbf{Z}\backslash\mathbf{E})$. For each $\mathbf{e} \in \mathfrak{X}_{\mathbf{E}}$, let $\mathbf{z}(\mathbf{e})$ agree with $\mathbf{z}_0$ outside $\mathbf{E}$ and

equal $\mathbf{e}$ on $\mathbf{E}$, and let $\mathbf{s}'(\mathbf{e})$ be the assignment to $\mathbf{S}'$ induced by $(\mathbf{x}, \mathbf{z}(\mathbf{e}))$. Set

$$A_{\mathbf{e}} = \mathbf{Q}(\mathbf{W}|\mathbf{U}, \mathbf{s}'(\mathbf{e})), \quad B_{\mathbf{e}} = \mathbf{Q}(\mathbf{W}|\mathbf{T}, \mathbf{x}, \mathbf{f}, \mathbf{e}), \quad H_{\mathbf{e}} = \mathbf{Q}(\mathbf{U}|\mathbf{T}, \mathbf{x}, \mathbf{f}, \mathbf{e}).$$

Also let $h = \mathbf{Q}(\mathbf{y}|\mathbf{U}, \mathbf{x})$, $q = \mathbf{Q}(\mathbf{U})$, $d = \mathbf{Q}_{\mathbf{s}'(\mathbf{e}_0)}(\mathbf{W})$, and $C_{\mathbf{e}} = \mathbf{Q}(\mathbf{y}|\mathbf{T}, \mathbf{x}, \mathbf{f}, \mathbf{e})$. Only $\mathbf{T} = \mathbf{Z} \cap \mathbf{S}$ is free in each block; the variables in $\mathbf{X} \cap \mathbf{S}$ remain fixed at $\mathbf{x}$. Condition (1), Condition (2) with the chain rule, corrected Condition (3) with the chain rule, and Condition (4) give, respectively,

$$Q_{\mathbf{x}}(\mathbf{y}) = hq, \tag{14}$$
$$C_{\mathbf{e}} = hH_{\mathbf{e}}, \tag{15}$$
$$B_{\mathbf{e}} = A_{\mathbf{e}}H_{\mathbf{e}}, \tag{16}$$
$$d = A_{\mathbf{e}_0}q. \tag{17}$$

The corrected invertibility requirements in (5)–(6) imply

$$H_{\mathbf{e}} = A_{\mathbf{e}}^{-1}B_{\mathbf{e}}, \tag{18}$$
$$q = A_{\mathbf{e}_0}^{-1}d. \tag{19}$$

Concatenating blocks in the same order over $\mathbf{e} \in \mathfrak{X}_{\mathbf{E}}$ gives

$$H = [A_{\mathbf{e}}^{-1}B_{\mathbf{e}}]_{\mathbf{e}} = \mathbf{Q}(\mathbf{U}|\mathbf{z}_0/(\mathbf{S} \cup \mathbf{Z}'), \mathbf{x}), \tag{20}$$
$$h = CH^{-1}, \qquad C = [C_{\mathbf{e}}]_{\mathbf{e}} = \mathbf{Q}(\mathbf{y}|\mathbf{z}_0/(\mathbf{S} \cup \mathbf{Z}'), \mathbf{x}). \tag{21}$$

Here H is invertible by (6). Substituting into Eq. (14) yields

$$Q_{\mathbf{x}}(\mathbf{y}) = C[A_{\mathbf{e}}^{-1}B_{\mathbf{e}}]_{\mathbf{e}}^{-1}A_{\mathbf{e}_0}^{-1}d.$$

For $\mathbf{Z}' = \emptyset$, the concatenation has one block and $B = AH$ is invertible. Hence the formula reduces to

$$Q_{\mathbf{x}}(\mathbf{y}) = \mathbf{Q}(\mathbf{y}|\mathbf{z}_0/\mathbf{S}, \mathbf{x})\mathbf{Q}(\mathbf{W}|\mathbf{z}_0/\mathbf{S}, \mathbf{x})^{-1}\mathbf{Q}_{\mathbf{s}'}(\mathbf{W}).$$

When $\mathbf{Z}' \neq \emptyset$, every $A_{\mathbf{e}}$ is supplied by the external study required in (6). In either case the final expression uses only available distributions. □

D Pseudoinverse and Intermediary Criteria

Lemma 4 (Chain-Rule Intermediary Criterion)**.** *Given a causal diagram $\mathcal{G}$, let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$, and $\mathbf{R}$ be disjoint subsets of $\mathbf{V}$ with $\mathbf{D}$ and $\mathbf{R}$ can be empty. Let $\mathbf{B} = \mathbf{B}' \dot\cup \mathbf{B}''$ and $\mathbf{C} = \mathbf{C}' \dot\cup \mathbf{C}''$ where $\mathbf{B}'$ and $\mathbf{C}'$ are not empty. Given an interventional distribution $Q = P_{\mathbf{r}}$, if $(\mathbf{C}' \perp\!\!\!\perp \mathbf{A} \mid \mathbf{B}, \mathbf{C}'', \mathbf{D})$ in $\mathcal{G}\backslash\mathbf{R}$ and $Q(\mathbf{a}, \mathbf{b}'', \mathbf{c}''|\mathbf{d}) = \sum_{\mathbf{b}', \mathbf{c}'} Q(\mathbf{a}|\mathbf{b}, \mathbf{c}, \mathbf{d})Q(\mathbf{b}|\mathbf{c}, \mathbf{d})Q(\mathbf{c}|\mathbf{d})$, then,*

$$\mathbf{Q}(\mathbf{A}, \mathbf{b}'', \mathbf{c}''|\mathbf{d}) = \mathbf{Q}(\mathbf{A}, \mathbf{b}''|\mathbf{C}', \mathbf{c}'', \mathbf{d}) \cdot \mathbf{Q}(\mathbf{B}', \mathbf{b}''|\mathbf{C}', \mathbf{c}'', \mathbf{d})^{\dagger} \cdot \mathbf{Q}(\mathbf{B}', \mathbf{b}'', \mathbf{c}''|\mathbf{d}).$$

Proof. For any fixed values $\mathbf{b}''$, $\mathbf{c}''$, and $\mathbf{d}$,

$$\begin{aligned} &Q(\mathbf{a}, \mathbf{b}'', \mathbf{c}''|\mathbf{d}) \\ &\quad = \mathbf{Q}(\mathbf{a}|\mathbf{B}', \mathbf{b}'', \mathbf{c}'', \mathbf{d})\mathbf{Q}(\mathbf{B}', \mathbf{b}''|\mathbf{C}', \mathbf{c}'', \mathbf{d})\mathbf{Q}(\mathbf{C}', \mathbf{c}''|\mathbf{d}) \\ &\quad = \mathbf{Q}(\mathbf{a}, \mathbf{b}''|\mathbf{C}', \mathbf{c}'', \mathbf{d})\mathbf{Q}(\mathbf{B}', \mathbf{b}''|\mathbf{C}', \mathbf{c}'', \mathbf{d})^{\dagger}\mathbf{Q}(\mathbf{B}', \mathbf{b}'', \mathbf{c}''|\mathbf{d}). \end{aligned}$$

□

We provide a visualization of the c-factorization intermediary criterion in Fig. 10. Given an unknown $P_{\mathbf{X}_\ell}(\mathbf{Y}_\ell)$ (top left) and known distributions (the three distributions at the bottom with $P_{\mathbf{x}_j}(\mathbf{y}_j)$ also shown at the top), we can focus on where a few variables are fixed to form a proper three-matrix multiplication. By replacing the submatrix of intermediary distribution based on the property of pseudoinverse, we can substitute the first two and the last two terms with submatrices that can be computed based on two non-intermediary distributions (the first and third distributions at the bottom.)

Theorem 3 (C-Factorization Intermediary Criterion)**.** *Let $\mathcal{G}$ be a causal diagram and $Q = P_{\mathbf{r}}$. Let $Q_{\mathbf{x}_\ell}(\mathbf{y}_\ell)$ be c-factorized as Eq.* (4)*. Let $\mathbf{X}_k^+$ be a subset of $\mathbf{X}_k$ excluding the rest five sets, $\{\mathbf{Y}_i, \mathbf{Y}_j, \mathbf{Y}_k, \mathbf{X}_i, \mathbf{X}_j\}$. $\mathbf{Y}_i^+$ is similarly defined. If $\mathbf{Z} \subseteq (\mathbf{X}_i \cap \mathbf{Y}_j)\backslash\mathbf{X}_k$ and $\mathbf{W} \subseteq (\mathbf{X}_j \cap \mathbf{Y}_k)\backslash\mathbf{X}_i$, then $\mathbf{Q}_{\mathbf{x}_\ell/\mathbf{X}_k^+}(\mathbf{y}_\ell/\mathbf{Y}_i^+)$, a submatrix of $\mathbf{Q}_{\mathbf{X}_\ell}(\mathbf{Y}_\ell)$, becomes*

$$\mathbf{Q}_{\mathbf{x}_\ell/\mathbf{X}_k^+}(\mathbf{y}_\ell/\mathbf{Y}_i^+) = \mathbf{Q}_{(\mathbf{x}_{ij}\backslash\mathbf{y}_{ij})/\mathbf{W}}((\mathbf{y}_{ij}\backslash\mathbf{Z})/\mathbf{Y}_i^+) \cdot \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z})^{\dagger} \cdot \mathbf{Q}_{(\mathbf{x}_{jk}\backslash\mathbf{y}_{jk})/\mathbf{X}_k^+}((\mathbf{y}_{jk}\backslash\mathbf{W})/\mathbf{Z}).$$

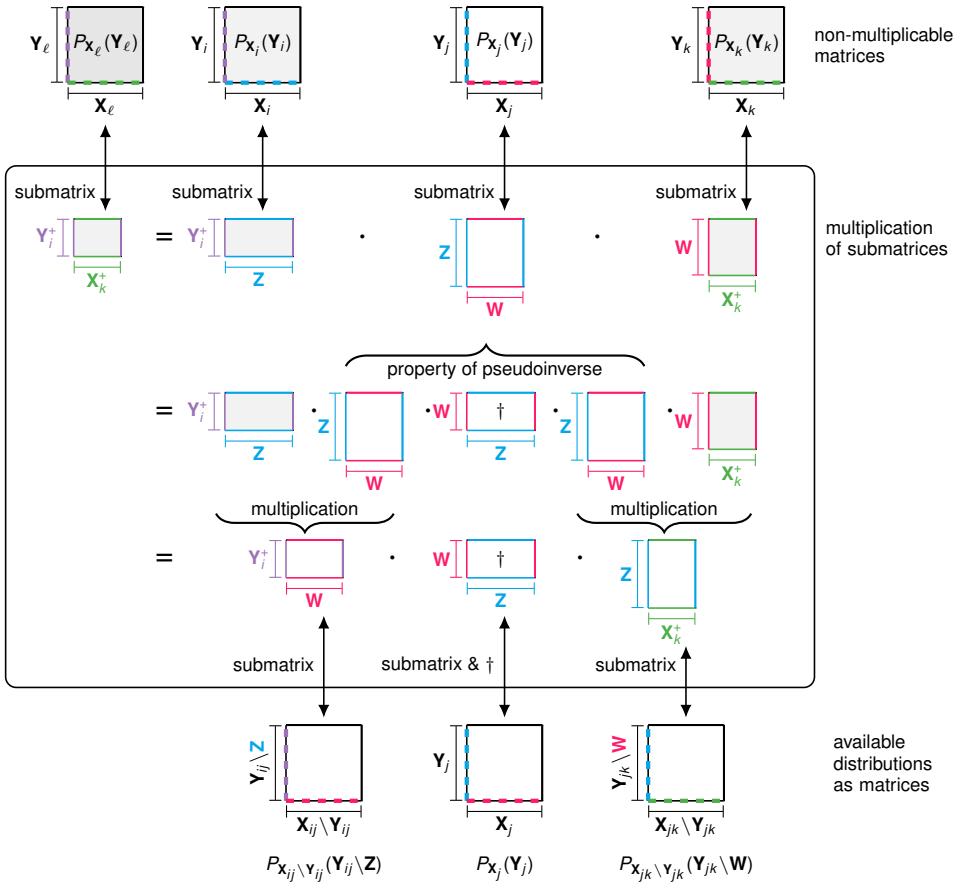


Figure 10: Illustration of the derivation of the factorization-based intermediary criterion.

Proof. Since $\mathbf{W} \subseteq \mathbf{Y}_k$ and $\mathbf{W} \cap \mathbf{X}_i = \emptyset$, the first factor is independent of the summation variables $\mathbf{W}$. Given the c-factorization of the query,

$$\begin{aligned} Q_{\mathbf{x}_\ell}(\mathbf{y}_\ell) &= \sum_{\mathbf{z}} Q_{\mathbf{x}_i}(\mathbf{y}_i) \sum_{\mathbf{w}} Q_{\mathbf{x}_j}(\mathbf{y}_j) Q_{\mathbf{x}_k}(\mathbf{y}_k) \\ &= \sum_{\mathbf{z}} \mathbf{Q}_{\mathbf{x}_i}(\mathbf{y}_i) \sum_{\mathbf{w}} \mathbf{Q}_{\mathbf{x}_j}(\mathbf{y}_j) \mathbf{Q}_{\mathbf{x}_k}(\mathbf{y}_k) \end{aligned}$$

The summation over $\mathbf{w}$ can be represented using matrix multiplication,

$$= \sum_{\mathbf{z}} \mathbf{Q}_{\mathbf{x}_i}(\mathbf{y}_i) \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j) \mathbf{Q}_{\mathbf{x}_k}(\mathbf{y}_k/\mathbf{W}).$$

If we consider only multiplying the first term, $\mathbf{Z}$ is allowed to be the subset of $\mathbf{Y}_j \cap \mathbf{X}_i$. However, we want to restrict $\mathbf{Z}$ does not affect the last matrix so as for us to

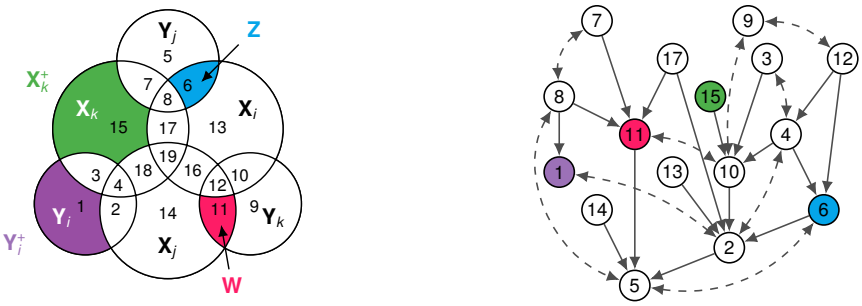
$$= \Big(\sum_{\mathbf{z}} \mathbf{Q}_{\mathbf{x}_i}(\mathbf{y}_i) \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j) \Big) \mathbf{Q}_{\mathbf{x}_k}(\mathbf{y}_k/\mathbf{W}).$$

This is possible when we exclude only $\mathbf{X}_k$ from $\mathbf{Y}_j \cap \mathbf{X}_i$ as $\mathbf{Y}_k$ is already disjoint to $\mathbf{Y}_j \cap \mathbf{X}_i$. Hence, with $\mathbf{Z} \subseteq (\mathbf{Y}_j \cap \mathbf{X}_i) \backslash \mathbf{X}_k$, the above expression is valid. Then, we proceed to rewrite it as,

$$\begin{aligned} &= \Big(\mathbf{Q}_{\mathbf{x}_i/\mathbf{Z}}(\mathbf{y}_i) \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z}) \Big) \mathbf{Q}_{\mathbf{x}_k}(\mathbf{y}_k/\mathbf{W}) \\ &= \mathbf{Q}_{\mathbf{x}_i/\mathbf{Z}}(\mathbf{y}_i) \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z}) \mathbf{Q}_{\mathbf{x}_k}(\mathbf{y}_k/\mathbf{W}). \end{aligned}$$

This results in the scalar $Q_{\mathbf{x}_\ell}(\mathbf{y}_\ell)$. We may enumerate $\mathbf{Y}_i^+$ as rows and $\mathbf{X}_k^+$ as columns because these variables do not occur in any other factor.

$$Q_{\mathbf{x}_\ell/\mathbf{X}_k^+}(\mathbf{y}_\ell/\mathbf{Y}_i^+) = \mathbf{Q}_{\mathbf{x}_i/\mathbf{Z}}(\mathbf{y}_i/\mathbf{Y}_i^+) \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z}) \mathbf{Q}_{\mathbf{x}_k/\mathbf{X}_k^+}(\mathbf{y}_k/\mathbf{W}).$$



(a) Euler diagram (b) a causal graph

Figure 11: (a) an Euler diagram depicting set relationships. (b) a causal graph where each number in (a) corresponds to a variable excluding 16, 18, and 19 for simplicity.

$$P_{13,14,15,17}(1\text{-}12) = P_{6,8,10,12,13,17}(1\text{-}4)P_{2,4,11,12,14}(5\text{-}8)P_{3,4,7,8,15,17}(9\text{-}12),$$

$$\begin{aligned} P_{13,14,15,17}(1,2,3,4,5,7,8,9,10,12) &= \sum_{6} P_{6,8,10,12,13,17}(1,2,3,4) \sum_{11} P_{2,4,11,12,14}(5,6,7,8)P_{3,4,7,8,15,17}(9,10,11,12) \\ &= \mathbf{P}_{V_6,8,10,12,13,17}(1,2,3,4)\mathbf{P}_{2,4,12,14,V_{11}}(5,V_6,7,8)\mathbf{P}_{3,4,7,8,15,17}(9,10,V_{11},12). \end{aligned}$$

Enumerating 1 and 15 gives

$$\mathbf{P}_{13,14,V_{15},17}(V_1,2,3,4,5,7,8,9,10,12) = \mathbf{P}_{V_6,8,10,12,13,17}(V_1,2,3,4)\mathbf{P}_{2,4,12,14,V_{11}}(5,V_6,7,8)\mathbf{P}_{3,4,7,8,V_{15},17}(9,10,V_{11},12).$$

Figure 12: Matrix equations for Fig. 11b, with $\mathbf{Z} = \{6\}$ and $\mathbf{W} = \{11\}$. Free row and column variables carry a V prefix; numbers without this prefix denote fixed values. Variable 12 is fixed in every matrix.

Now, we apply matrix multiplication of c-factors (Lemma 2) to examine the matrix multiplication of the first two terms and the last two terms.

$$\begin{aligned} \mathbf{Q}_{\mathbf{x}_i/\mathbf{Z}}(\mathbf{y}_i/\mathbf{Y}_i^+)\mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z}) &= \mathbf{Q}_{(\mathbf{x}_{ij}\setminus\mathbf{y}_{ij})/\mathbf{W}}((\mathbf{y}_{ij}\setminus\mathbf{Z})/\mathbf{Y}_i^+) \\ \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z})\mathbf{Q}_{\mathbf{x}_k/\mathbf{X}_k^+}(\mathbf{y}_k/\mathbf{W}) &= \mathbf{Q}_{(\mathbf{x}_{jk}\setminus\mathbf{y}_{jk})/\mathbf{X}_k^+}((\mathbf{y}_{jk}\setminus\mathbf{W})/\mathbf{Z}). \end{aligned}$$

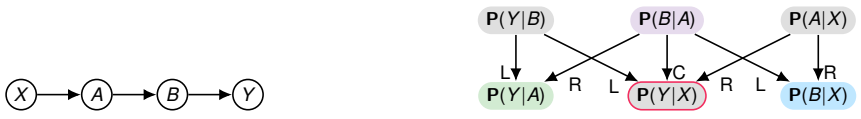
Plugging the last two expressions into the base intermediary criterion (Lemma 3), we finally yield

$$\mathbf{Q}_{(\mathbf{x}_{ij}\setminus\mathbf{y}_{ij})/\mathbf{W}}((\mathbf{y}_{ij}\setminus\mathbf{Z})/\mathbf{Y}_i^+) \cdot \mathbf{Q}_{\mathbf{x}_j/\mathbf{W}}(\mathbf{y}_j/\mathbf{Z})^\dagger \cdot \mathbf{Q}_{(\mathbf{x}_{jk}\setminus\mathbf{y}_{jk})/\mathbf{X}_k^+}((\mathbf{y}_{jk}\setminus\mathbf{W})/\mathbf{Z}).$$

□

We present a matrix equation for Fig. 11b in Fig. 12, with $\mathbf{Z} = \{6\}$ and $\mathbf{W} = \{11\}$. The value of variable 12 is retained and fixed consistently across all three factors: since $12 \in \mathbf{X}_i$, it cannot be included in $\mathbf{W}$. The free row and column variables are 1 and 15, respectively. Each numbered variable may also represent a set of variables.

We provide a schematic of simple intermediary criterion example Fig. 13a in Fig. 13b where $\mathbf{P}(Y|X)$ is the result of three-matrix multiplications. This example corresponds to the counterexample used in Prop. 1.



(a) (b)

Figure 13: (a) A causal diagram demonstrating the identification of $P_x(y) = P(y|x)$ given $P(B|x)$, $P(B|A)$, and $P(y|A)$. (b) A schematic diagram for the example (Fig. 13a) where the pseudoinverse of $\mathbf{P}(B|A)$ is a key to identifying $P(y|x) = P_x(y)$. This involves the multiplication of three matrices with the order annotated with L, C, and R.

E A Unifying Causal Identification Algorithm

Theorem 4. *ID-ME is sound.*

Proof. Maintain the following invariant: every distribution in the expanded input and every factor recorded in $\mathbb{F}'$ has an expression in the original inputs equal to that distribution in every compatible model satisfying the stated rank and support conditions. The invariant initially holds. The query reduction in Line 2 is valid by Rule 3 of do-calculus. The corrected projection range in Line 4 ensures that every enumerated c-factorization is defined.

In Line 3, marginalization, conditioning on the stated support, and do-calculus preserve the invariant. Matrix inversion steps are applications of Lemma 1 only when the inverse of the required orientation exists; intermediary steps are justified by Lemma 4. In Lines 6–8, successful ID-RC calls are sound on the original graph by [4, 11], and successful PROXY calls are sound by the corrected Thms 1 and 2 and the rank-form derivation of Eq. (1) in Appendix C, including the applicable rank, per-slice invertibility and support conditions. In Line 9, CF-INV uses Lemma 2 with its applicable inverse condition, and CF-INT uses Thm. 3 with both set conditions. Each call checks that its factors belong to the required common graphical factorization and have consistent assignments. Thus each successful insertion preserves the invariant.

Finally, exact-cover returns only a combination that satisfies the c-factorization identity Eq. (2) in an eligible projection. Substituting the valid expressions recorded in $\mathbb{F}'$ into that identity yields the reduced query. By Line 2 it also yields the original query. Consequently every returned formula is correct. □

Theorem 5. *ID-ME strictly subsumes proxy criteria [8, 14], GID(-PO) [13, 12], mID, or eID [11].*

Proof. For unconditional inputs, eID reduces to mID: the conditional chain-rule closure of Definition 6 in Lee and Shpitser [11] is vacuous when every conditioning set is empty. The mID condition corresponds to GID-PO, with GID as the nonmarginal special case [13, 12, 11]. If the inputs are both unconditional and nonmarginal, completeness of GID supplies the ordinary factorization used by the first branch of Line 4; ID-RC recovers those factors.

In the remaining marginal or conditional cases, consider any factorization accepted by mID or eID. When conditional inputs are present, expand retains the chain-rule closure required by eID. For each candidate factor, ID-RC applies ordinary single-input recovery for an unconditional input and Lemma 5 of Lee and Shpitser [11] for a conditional input. To see that Line 4 includes its projection, let $\mathbf{Y}'$ be any candidate outcome set accepted by Lemma 6 of Lee and Shpitser [11]. That lemma restricts $\mathbf{Y} \subseteq \mathbf{Y}'$ to ancestors of $\mathbf{Y}$ after the treatment intervention. Thus $\mathbf{X} \cup \mathbf{Y}' \subseteq \mathrm{An}^{\mathcal{G}}(\mathbf{Y})$ after the reduction in Line 2. Put $A = \mathrm{An}^{\mathcal{G}}(\mathbf{Y})$. Since A is ancestral, a latent-projection path between its vertices cannot pass through a nonancestor; hence $\mathcal{G}\langle \mathbf{X} \cup \mathbf{Y}' \rangle = \mathcal{G}[A]\langle \mathbf{X} \cup \mathbf{Y}' \rangle$. This projection is explicitly enumerated in Line 4. Its target c-factors are exactly the districts identified in that successful eID factorization. The original graph remains available to expand and ID-RC for processing input distributions. Finally, exact-cover can select the recovered factors of the accepted factorization. Thus ID-ME returns an identifying expression whenever any of these graphical procedures does.

For the original single- and double-proxy criteria [8, 14], the unconditional nonmarginal case is covered by the preceding GID argument. Otherwise, the single-outcome query occurs among the c-factors enumerated in the projection onto its retained treatments and outcome. The PROXY module applies the generalized criteria to this factor. If $|\mathfrak{X}_W| = |\mathfrak{X}_Z| = |\mathfrak{X}_U|$, the original MGT case is Thm. 2 with $\mathbf{S} = \mathbf{X} \cup \mathbf{Z}$ and $\mathbf{Z}' = \emptyset$, so the extra per-slice requirement is vacuous. Otherwise, PROXY uses the rank form of Eq. (1) under (C1)–(C4), whose derivation in Appendix C allows rectangular proxy matrices.

The chain example in Prop. 1 satisfies the corrected intermediary conditions with factor order $(B, Y), (A, B), (X, A)$, $\mathbf{Z} = \{B\}$, and $\mathbf{W} = \{A\}$. Its expression $\mathbf{P}(y|A)\mathbf{P}(B|A)^{\dagger}\mathbf{P}(B|x)$ is outside the graphical procedures above. Since A and B are mediators, the only adjustment-admissible choice of U among the remaining vertices is empty. No nonempty proxy then satisfies the original independence requirements for this query; the degenerate empty-proxy case would require the unavailable $P(Y|X)$. Thus the chain example alone establishes strictness over the named union. The

relaxed proxy examples in Figs. 3a to 3e, with the stated rank and support conditions, additionally give cases outside the original proxy templates. □

Changes

Unofficial AI-corrected version · 2026-10-25. Sanghack Lee and Elias Bareinboim, *Causal Identification with Matrix Equations*, NeurIPS 2021; technical report R-70, August 2022. Original: `https://causalai.net/r70.pdf`. This version: `https://www.sanghacklee.me/ai-corrected/matrix_equations_AI_corrected_2026-10-25.pdf`.

These proposed corrections were prepared with AI assistance. **Author review pending.** This document is not a publisher-issued correction.

Location	Correction and assumptions
Theorem 1; Eq. (1); Appendix C	Full-column-rank inversions in Theorem 1 use Moore–Penrose pseudoinverses. The elementwise division is restricted to $\operatorname{supp} Q(\mathbf{U})$, with $Q(\mathbf{u}\|\mathbf{x}) > 0$ whenever $Q(\mathbf{u}) > 0$. Eq. (1) and its motivational applications use pseudoinverses under the original rank-$\|\mathfrak{X}_U\|$ condition; explicitly square-invertible examples retain ordinary inverses.
Theorem 2; Appendix C	Condition (3) reads $(\mathbf{W} \perp\!\!\!\perp \mathbf{S} \mid \mathbf{U}, \mathbf{S}')$, with $\mathbf{S}' = (\mathbf{X} \cup \mathbf{Z}) \backslash \mathbf{S}$. When $\mathbf{Z}' \neq \emptyset$, each slice of $Q(\mathbf{W}\|\mathbf{U}, \mathbf{s}'/\mathbf{Z}')$ indexed by $\mathbf{z}' \in \mathfrak{X}_{\mathbf{Z}'}$ must be invertible. The proof distinguishes the varying slice used for reconstruction from the selected slice used with the surrogate distribution.
Theorem 3; Appendix D; Figs. 11–12	For the displayed ordering, add $\mathbf{W} \cap \mathbf{X}_i = \emptyset$ to the existing restriction $\mathbf{Z} \cap \mathbf{X}_k = \emptyset$. The example uses $\mathbf{W} = \{11\}$ and retains variable 12 at a consistent fixed value; the red region in the Euler diagram matches this restriction.
Algorithm 1; Theorems 4–5	Line 4 projects only onto subsets of $\mathrm{An}_{\mathcal{G}}(\mathbf{Y})$. The modules enforce the corrected criteria; PROXY also applies Eq. (1)'s rank form to rectangular proxy matrices. The soundness proof states the composition invariant. The inclusion proof covers the original proxy criteria, GID(-PO), mID and eID in the discrete setting, with the chain example establishing strictness. Theorem statements 4 and 5 are retained.
Section 4.1; appendices and figures	The criterion for Figure 3b uses $P(W\|U, x, Y)$ and $P_{x,y}(W)$. Retain the common intervention $\mathbf{x}_i \cap \mathbf{x}_j$ in the two-factor lemma's proof. Correct the adjustment criterion's $W \in \mathbf{X}$ to $W \notin \mathbf{X}$, do-calculus graph notation, the Theorem-3 proof's set difference, and schematic operand labels. Clarify slice notation and latent-DAG separation; group typographic, cross-reference and citation repairs.

The paper reports no experiments. Finite-model calculations provide supplementary checks of the corrected formulae; the mathematical claims rely on the stated assumptions and proofs.

Please cite the original paper. When this correction is used, note its date separately in the same reference.

AI review sign-off: claude-opus-5-5 · 2026-10-25.